Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering &Technology

ESTD:1995

Department of Computer Science & Engineering

SUBJECT: Computer Networks

Subject Code:19PC1CS07

III B.Tech- I Semester- A

Dr. MUNIRAJU NAIDU VADLAMUDI

Assistant Professor

Email: munirajunaidu_v@vnrvjiet.in

Syllabus

ESTD:1995

UNIT – I:

- Data Communication Components: Representation of data and its flow Networks, Various Connection Topology, Protocols and Standards, OSI model, Transmission Media, Overview of LAN: Wired LAN, Wireless LANs, Connecting LAN and Virtual LAN, Techniques for Band width utilization: Multiplexing - Frequency division, Time division and Wave division, Concepts on spread spectrum.

UNIT – II:

- Data Link Layer and Medium Access Sub Layer: Error Detection and Error Correction Fundamentals, Block coding, Hamming Distance, CRC; Flow Control and Error control protocols - Stop and Wait, Go back – N ARQ, Selective Repeat ARQ, Sliding Window, Piggybacking, Random Access, Multiple access protocols -Pure ALOHA, Slotted ALOHA, CSMA/CD, CDMA/CA

UNIT – III:

- Network Layer: Switching, Logical addressing – IPV4, IPV6; Address mapping – ARP, RARP, BOOTP and DHCP–Delivery, Forwarding and Unicast Routing protocols.

Syllabus

ESTD:1995

UNIT – IV:

- Transport Layer: Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), SCTP Congestion Control; Quality of Service, QoS improving techniques: Leaky Bucket and Token Bucket algorithm.

UNIT – V:

- Application Layer: Domain Name Space (DNS), DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Firewalls.

UNIT – VI:

- Security: Cryptography- Symmetric-Key Cryptography, Asymmetric-Key Cryptography, Network Security-Security Services, Message Confidentiality, Message Integrity, Message Authentication, Digital Signature, Entity Authentication, Key Management.
- **TEXT BOOKS:**
- 1. Data Communication and Networking, Behrouz A. Forouzan, 4th Edition, Tata McGraw-Hill
- 2. Computer Networks, Andrew S. Tanenbaum, 8th Edition, Pearson New International Edition
- **REFERENCES:**
- 1. Data and Computer Communication, William Stallings, 8th Edition, Pearson Prentice Hall India
- 2. Internetworking with TCP/IP, Volume 1, Douglas Comer, 6th Edition, Prentice Hall of India
- 3. TCP/IP Illustrated, Volume 1, W. Richard Stevens, Addison-Wesley

Agenda

ESTD:1995

- Representation of data and its flow Networks
- Various Connection Topology
- Protocols and Standards,
- OSI model,
- Transmission Media,
- Techniques for Bandwidth utilization:
  - Multiplexing
    - Frequency division
    - Time division

DATA COMMUNICATIONS

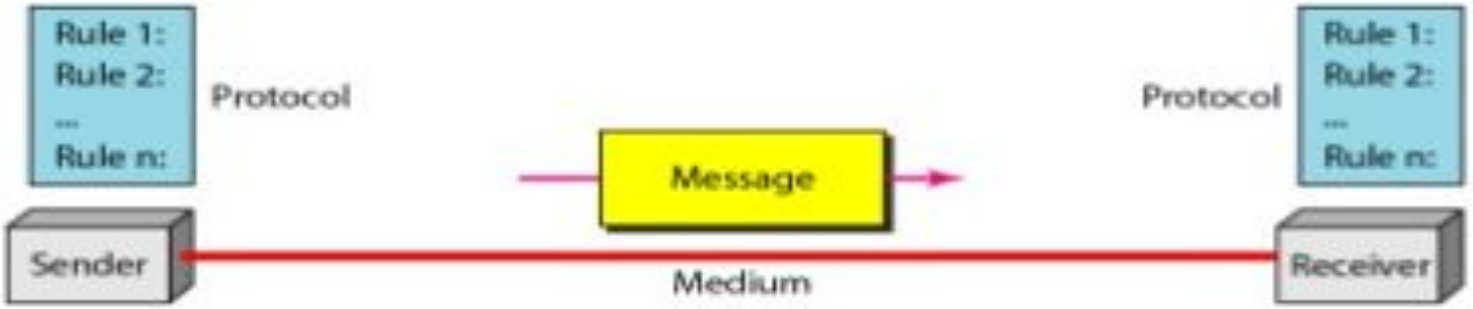
ESTD:1995

- When we communicate, we are sharing information.
- sharing can be local or remote.
- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).
- The effectiveness of a data communications system depends on four fundamental characteristics:
  - Delivery: To the correct destination
  - Accuracy: data delivered accurately , not altered during transmission
  - Timeliness
- Jitter: Jitter refers to the variation in the packet arrival time. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result

Components

ESTD:1995

- A data communications system has five components



Message. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

Components

ESTD:1995

- **Receiver.** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
- **Transmission medium**. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
- **Protocol.** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Data Representation

ESTD:1995

Information comes in different forms such as ***text, numbers, images, audio, and video*.**

- **Text:** In data communications, text is represented as a bit pattern, a sequence of bits (0s or 1s). Unicode, uses 32 bits to represent a symbol or character used in any language in the world. (ASCII) constitutes the first 127 characters in Unicode and is also referred to as Basic Latin.
- **Numbers** are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations.
- **Images** are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the *resolution.*
- **Audio** refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete.

Data Representation

ESTD:1995

Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion

Data Flow

ESTD:1995

Communication between two devices can be simplex, half-duplex, or full-duplex as shown in the below Figure .



Data Flow

ESTD:1995

- **Simplex:** In simplex mode, the communication is unidirectional. Only one of the two devices on a link can transmit; the other can only receive.

Example: Radio Station

- **Half-Duplex:** In half-duplex mode, each station can both transmit and receive, but not at the same time. : When one device is sending, the other can only receive, and vice versa. Examples: Walkie-Talkie,browsing,USB,Railway Tracks
- **Full-Duplex: In full-duplex mode**, both stations can transmit and receive simultaneously. **In full-duplex mode**, signals going in one direction share the capacity of the link with signals going in the other direction.
  - This sharing can occur in two ways:
    - Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or
    - the capacity of the channel is divided between signals traveling in both directions.
    - Example: Telephone Call

NETWORKS

ESTD:1995

- A network is a set of devices (referred to as *nodes)* connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
  - ***Distributed Processing:***
    - Most networks use distributed processing, in which a task is divided among multiple computers.
    - Instead of one single large machine being responsible for all aspects of a process, separate computers (usually a personal computer or workstation) handle a subset.
  - **Network Criteria:**
    - **A** network must be able to meet a certain number of criteria. The most important of these are ***performance, reliability, and security***.

ESTD:1995

Performance

- Performance can be measured using transit time and response time.
- *Transit time* is the amount of time required for a message to travel from one device to another.
- *Response time* is the elapsed time between an inquiry and a response
- The performance of a network depends on a number of factors, including the number of users, the type of transmission medium, hardware capabilities , and the efficiency of the software.

NETWORKS

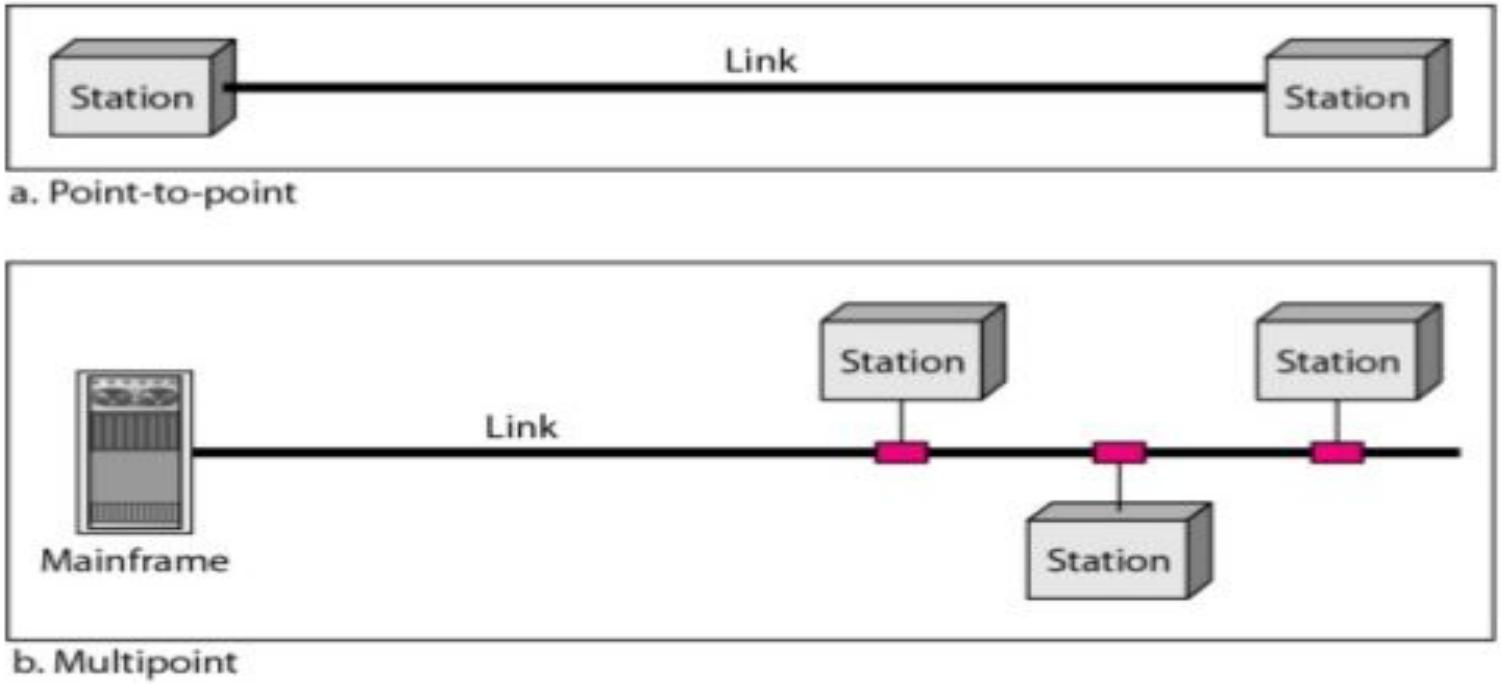
ESTD:1995

- Performance is evaluated by two networking metrics: ***throughput and delay*****.** More throughput and less delay. These two criteria are often contradictory. Sending more data to the network, throughput increased and hence delay also increased due to traffic congestion in the network
- **Reliability: In** addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.
- **Security:** Network security issues include protecting data from unauthorized access, damage and implementing policies and procedures for recovery from breaches and data losses.

Type of Connection

ESTD:1995

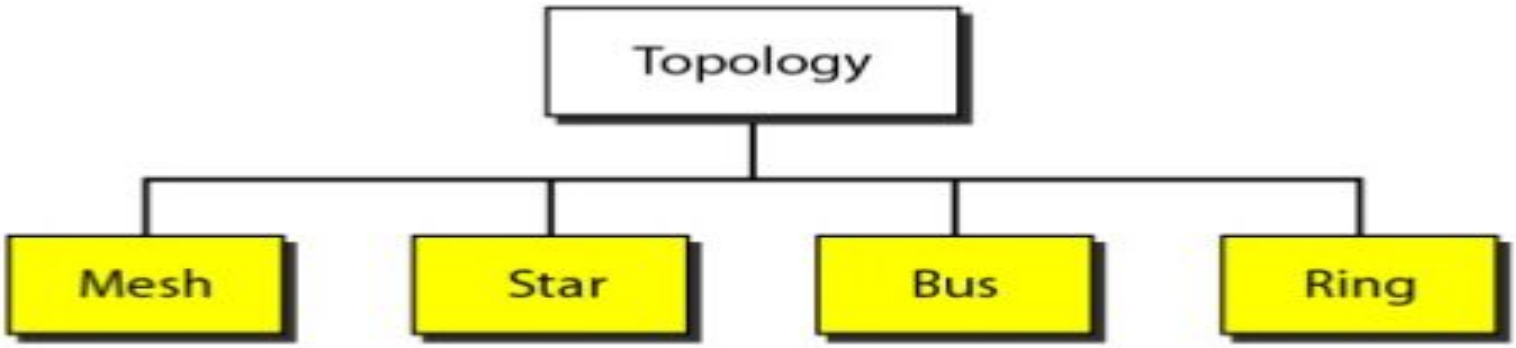
- A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another.
- ***There are two possible types of connections: point-to-point and multipoint.***



Type of Connection

ESTD:1995

The term *physical topology* refers to the way in which a network is laid out physically. Two or more devices connect to a link; ***two or more links form a topology.*** The *topology of a network is the geometric representation of the relationship of all the links and linking devices* (usually called nodes) to one another. ***There are four basic topologies possible: mesh, star, bus, and ring .***



Mesh topology

ESTD:1995

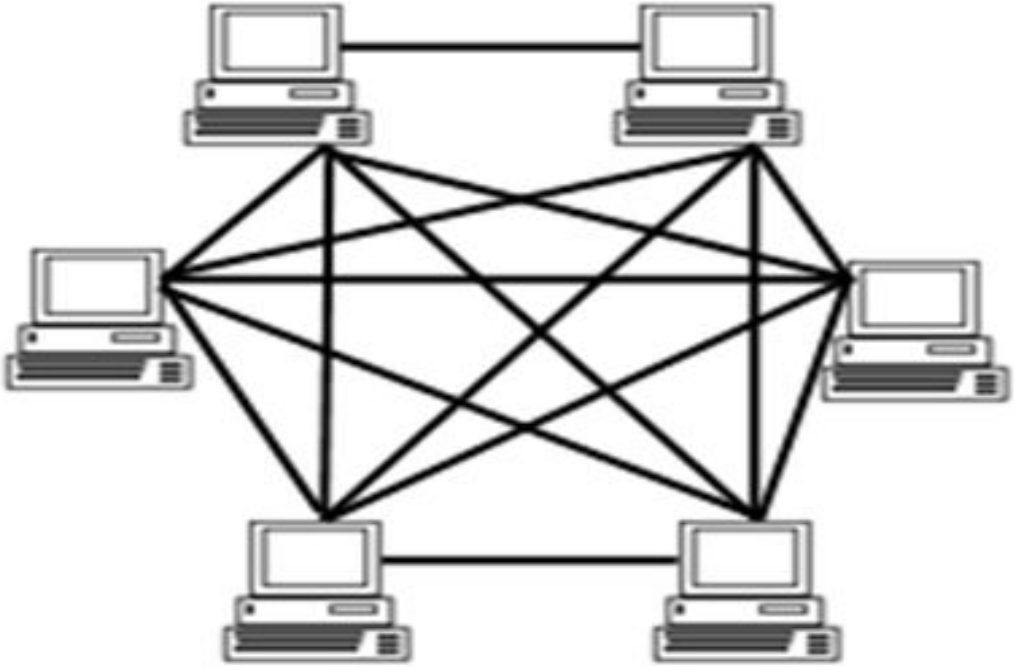
Mesh: In a mesh topology, every device has a dedicated point-to-point link to every other device.

The term *dedicated* means that the link carries traffic only between the two devices it connects.

To find the number of physical links in a fully connected mesh network with *n* nodes, we first consider that each node must be connected to every other node. Node 1 must be connected to *n* - 1 nodes, node 2 must be connected to *n-1* nodes, and finally node *n* must be connected to *n* - 1 nodes.

We need *n(n* - 1) physical links. However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology, we need $\frac{n(n-1)}{2}$ duplex –mode links.

Mesh topology



Advantages of Mesh topology

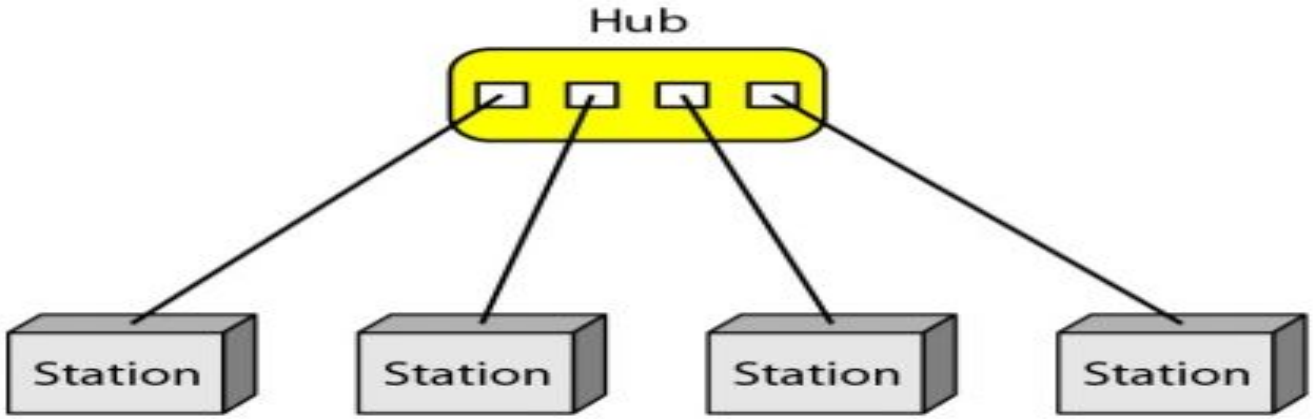
ESTD:1995

- ***Advantages of Mesh topology***
  - No data traffic issues as there is a dedicated link between two devices.
  - reliable and robust as failure of one link doesn't affect other links and the communication between other devices on the network.
  - It is secure because there is a point to point link thus unauthorized access is not possible.
  - Fault detection is easy.
- ***Disadvantages of Mesh topology***
  - Amount of wires required to connected each system is tedious and headache.
  - Since each device needs to be connected with other devices, number of I/O ports required must be huge.
  - Scalability issues because a device cannot be connected with large number of devices with a dedicated point to point link.

Star Topology

ESTD:1995

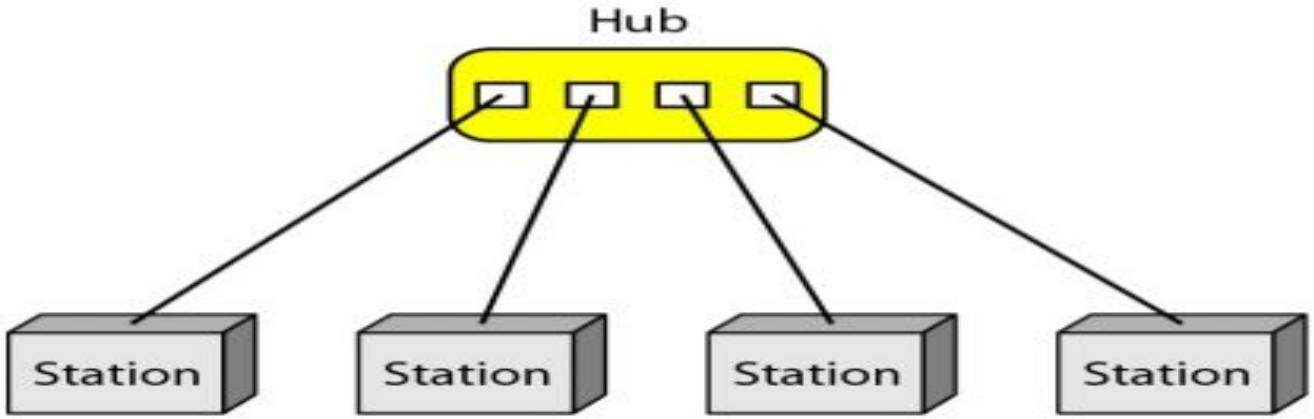
- In star topology each device in the network is connected to a central device called hub.
- star topology doesn't allow direct communication between devices, a device must have to communicate through hub.
- If one device wants to send data to other device, it has to first send the data to hub and then the hub transmit that data to the designated device.



Star Topology

ESTD:1995

- In star topology each device in the network is connected to a central device called hub.
- star topology doesn't allow direct communication between devices, a device must have to communicate through hub.
- If one device wants to send data to other device, it has to first send the data to hub and then the hub transmit that data to the designated device.



Star Topology

ESTD:1995

- **Advantages of Star topology**
  - Less expensive because each device only need one I/O port and needs to be connected with hub with one link.
  - Easier to install
  - Less amount of cables required because each device needs to be connected with the hub only.
  - Robust, if one link fails, other links will work just fine.
  - Easy fault detection because the link can be easily identified.
- **Disadvantages of Star topology**
  - If hub goes down everything goes down, none of the devices can work without hub.
  - Hub requires more resources and regular maintenance because it is the central system of star topology.

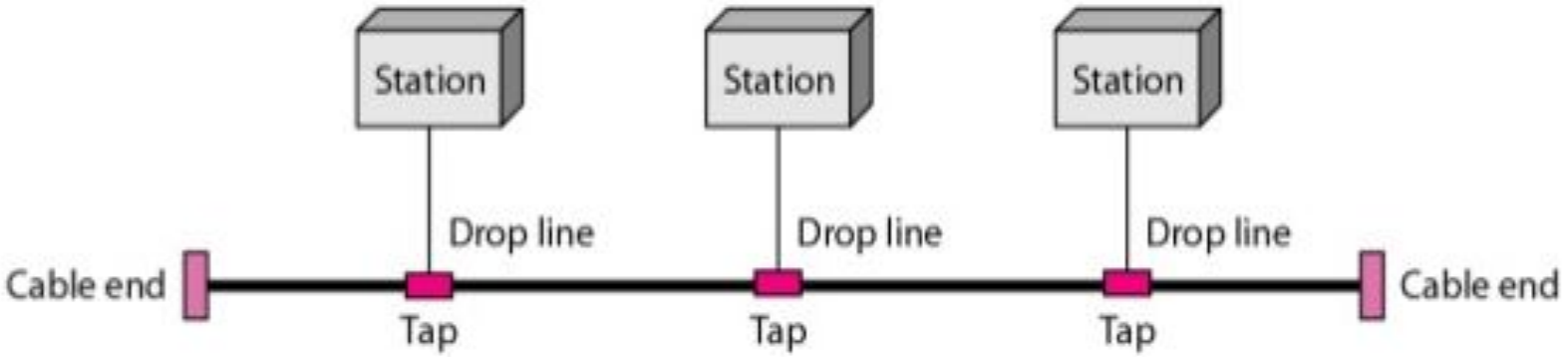
Star Topology

ESTD:1995

- **Advantages of Star topology**
  - Less expensive because each device only need one I/O port and needs to be connected with hub with one link.
  - Easier to install
  - Less amount of cables required because each device needs to be connected with the hub only.
  - Robust, if one link fails, other links will work just fine.
  - Easy fault detection because the link can be easily identified.
- **Disadvantages of Star topology**
  - If hub goes down everything goes down, none of the devices can work without hub.
  - Hub requires more resources and regular maintenance because it is the central system of star topology.

Bus Topology

Bus Topology: Mesh and star topologies are examples of point-to-point connections. Bus topology is a multi-point. One long cable acts as a **backbone** to link all the devices in a network .



Bus Topology

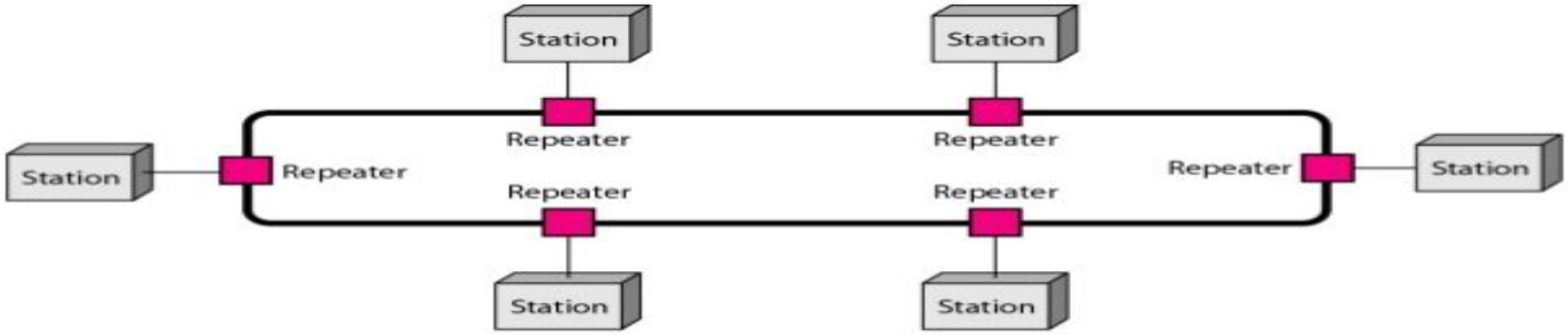
ESTD:1995

- ***Advantages of bus topology***
  - Easy installation, each cable needs to be connected with backbone cable.
  - Less cables required than Mesh and star topology
- ***Disadvantages of bus topology***
  - Difficultly in fault detection.
  - Not scalable as there is a limit of how many nodes you can connect with backbone cable.

Ring Topology

ESTD:1995

- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater.
- When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along



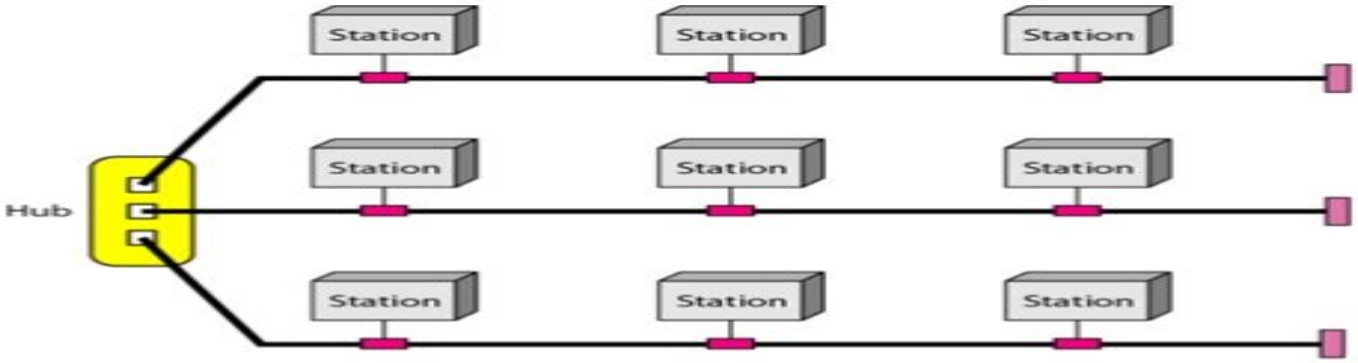
Ring Topology

ESTD:1995

- ***Advantages of Ring Topology***
  - Easy to install.
  - Managing is easier as to add or remove a device from the topology only two links are required to be changed.
- ***Disadvantages of Ring Topology***
  - A link failure can fail the entire network as the signal will not travel forward due to failure.
    Data traffic issues, since all the data is circulating in a ring.
  - Ring topology was prevalent when IBM introduced its local-area network Token Ring. Today, the need for higher-speed LANs has made this topology less popular.
- **Hybrid Topology** A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology

Hybrid Topology

ESTD:1995



Advantages of Hybrid topology

We can choose the topology based on the requirement for example, scalability is our concern then we can use star topology instead of bus technology.

Scalable as we can further connect other computer networks with the existing networks with different topologies.

Disadvantages of Hybrid topology

- Fault detection is difficult.
- Installation is difficult.
- Design is complex so maintenance is high thus expensive

Network Models

ESTD:1995

Network Models

- Computer networks are created by different entities. Standards are needed so that these heterogeneous networks can communicate with one another. The two best-known standards are the OSI model and the Internet model.

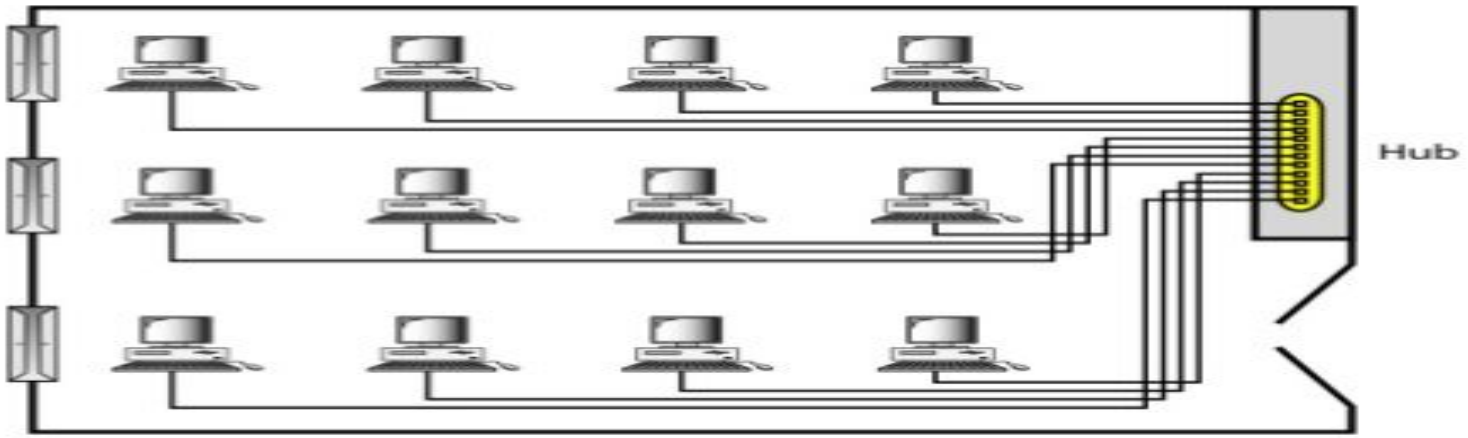
Categories of Networks

- There are two primary categories: local-area networks and wide-area networks. The category into which a network falls is determined by its size. A LAN normally covers an area less than 2 miles; a WAN can be worldwide. Networks of a size in between are normally referred to as metropolitan area networks and span tens of miles.

Network Models

ESTD:1995

- ***Local Area Network***
  - A local area network (LAN) is usually privately owned and links the devices in a single office, building, or campus.



LANs are designed to allow resources to be shared between personal computers or workstations. The resources to be shared can include hardware (e.g., a printer), software (e.g., an application program), or data.

Network Models

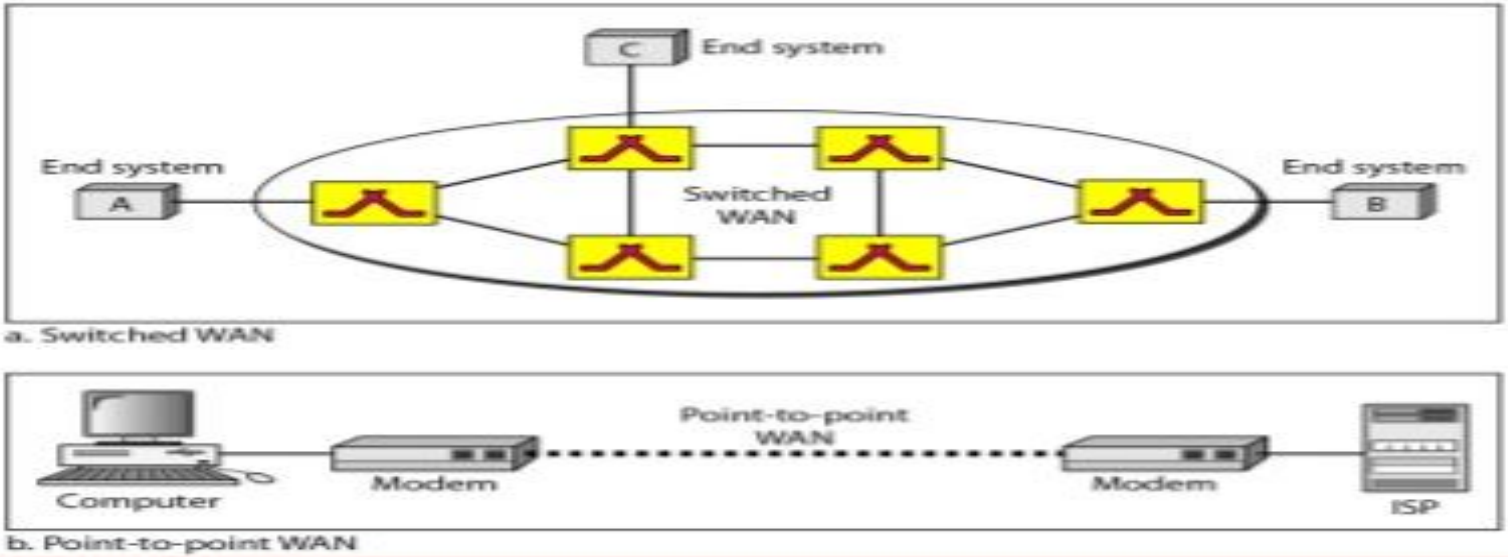
ESTD:1995

- LANs are distinguished from other types of networks by their transmission media and topology. In general, a given LAN will use only one type of transmission medium. The most common LAN topologies are bus, ring, and star.

❑ *Wide Area Network*

- A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent, or even the whole world.
- A WAN connects to the Internet or as simple as a dial-up line that connects a home computer to the Internet. The first is a switched WAN and to the second is a point-to-point WAN.

Network Models



- The switched WAN connects the end systems, which usually comprise a router (internetworking connecting device) that connects to another LAN or WAN.
- The point-to-point WAN is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (ISP). This type of WAN is often used to provide Internet access.

Network Models

ESTD:1995

- ***Metropolitan Area Networks***
  - A metropolitan area network (MAN) is a network with a size between a LAN and a WAN.
  - Normally covers the area inside a town or a city.
  - Designed for customers who need a high- speed connectivity. A good example of a MAN is the part of the telephone company network that can provide a high-speed DSL line to the customer.
- **Interconnection of Networks: Internetwork**
  - Today, it is very rare to see a LAN, a MAN, or a LAN in isolation; they are connected to one another. When two or more networks are connected, they become an internetwork, or internet.

Network Models

ESTD:1995

- An **intranet** is a series of networks belonging to one organization (i.e., a company), usually separated by building, cities, region, and or countries/continents. The important thing is that they are all owned by the same organization.
- An **internet** (small "i") is a series of networks owned by two or more organizations (Microsoft and Google, for example) and set up to communicate with each other.
- The **Internet** (big "I") is the worldwide network comprising of all other networks interconnected and communicating on the open Web. The IP addresses and DNS names are organized and issued by organization designated by the host country. This is what we refer to, public network that connects the globe

PROTOCOLS AND STANDARDS

ESTD:1995

- ***Protocols***
  - In computer networks, communication occurs between entities in different systems.
  - **A protocol is a set** of rules that govern data communications. A protocol defines what is communicated, how it is communicated, and when it is communicated. The key elements of a protocol are syntax, semantics, and timing.
    - **Syntax.** structure or format of the data, meaning the order in which they are presented.
    - **Semantics.** Meaning of each section of bits.
    - **Timing.** The term *timing* refers to **two characteristics**: when data should be sent and how fast they can be sent.

PROTOCOLS AND STANDARDS

ESTD:1995

- Standards are essential in creating and maintaining an open and competitive market for equipment manufacturers and in guaranteeing national and international interoperability of data and telecommunications technology and processes.
- Standards provide guidelines to manufacturers, vendors, government agencies, and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications.
- Data communication standards fall into two categories: *de facto* (meaning "by fact" or "by convention") and *de jure* (meaning "by law" or "by regulation").

PROTOCOLS AND STANDARDS

ESTD:1995

Standards Organizations

- International Organization for Standardization (ISO).
- International Telecommunication Union-Telecommunication Standards Sector (ITU-T).
- American National Standards Institute (ANSI).
- Institute of Electrical and Electronics Engineers (IEEE).
- Electronic Industries Association (EIA).

THE OSI MODEL

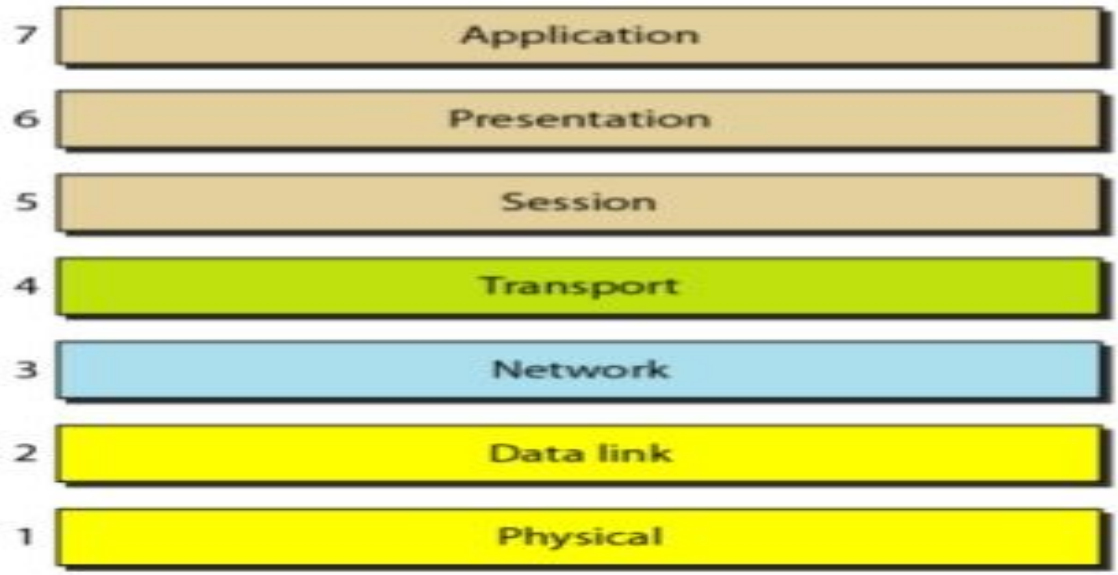
ESTD:1995

- **An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.**
- The *purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software*.
- The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.

THE OSI MODEL

ESTD:1995

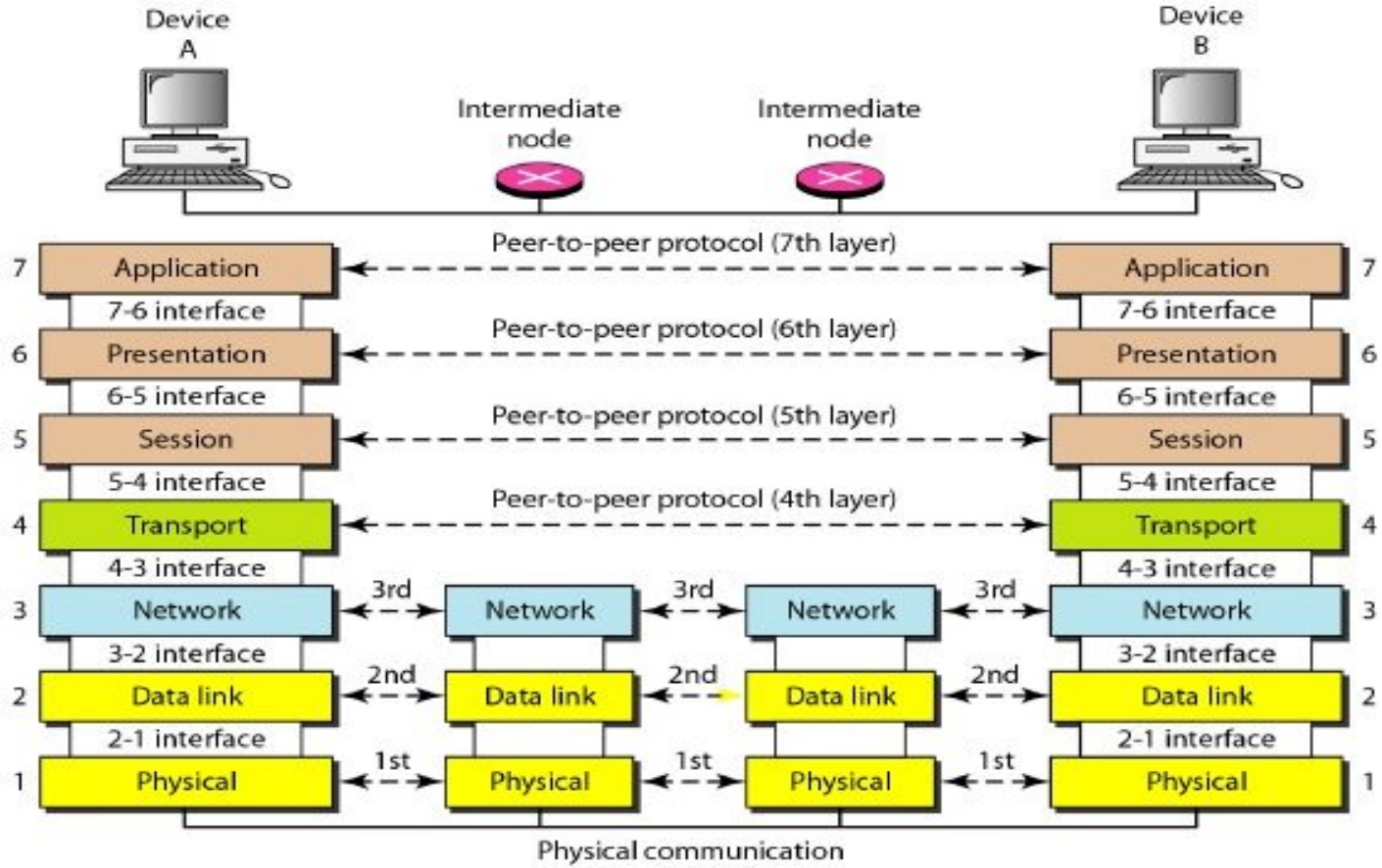
- **Layered Architecture**
  - The OSI model is composed of seven ordered layers: physical (layer 1), data link (layer 2), network (layer 3), transport (layer 4), session (layer 5), presentation (layer 6), and application (layer 7).



THE OSI MODEL

ESTD:1995

- **Peer-to-Peer Processes**



THE OSI MODEL

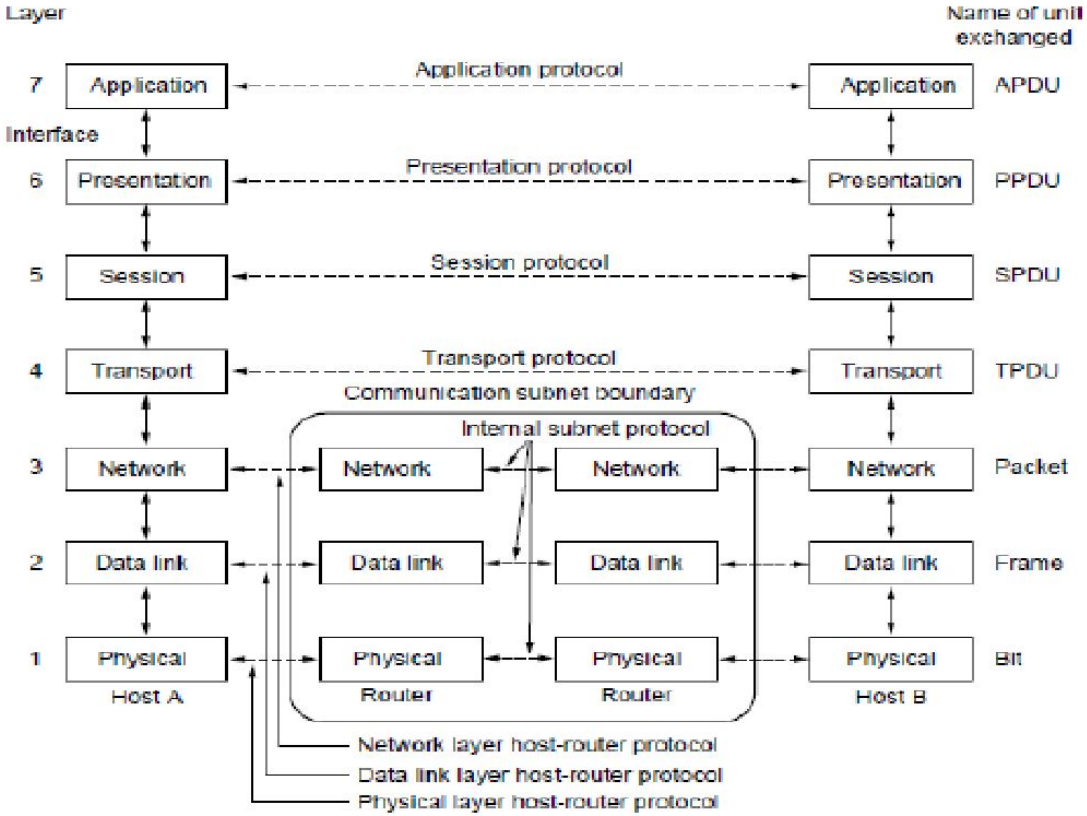
ESTD:1995

- ***Interfaces Between Layers***
  - Each interface defines the information and services a layer must provide for the layer above it. Well-defined interfaces and layer functions provide modularity to a network.
- ***Organization of the Layers***
  - The seven layers can be thought of as belonging to three subgroups. Layers I, 2, and 3-physical, data link, and network-are the network support layers; they deal with the physical aspects of moving data from one device to another (such as electrical specifications, physical connections, physical addressing, and transport timing and reliability).
  - Layers 5, 6, and 7-session, presentation, and application-can be thought of as the user support layers; they allow interoperability among unrelated software systems. Layer 4, the transport layer, links the two subgroups and ensures that what the lower layers have transmitted is in a form that the upper layers can use. The upper OSI layers are almost always implemented in software; **lower layers are a combination of hardware and software, except for the physical layer, which is mostly hardware.**

THE OSI MODEL

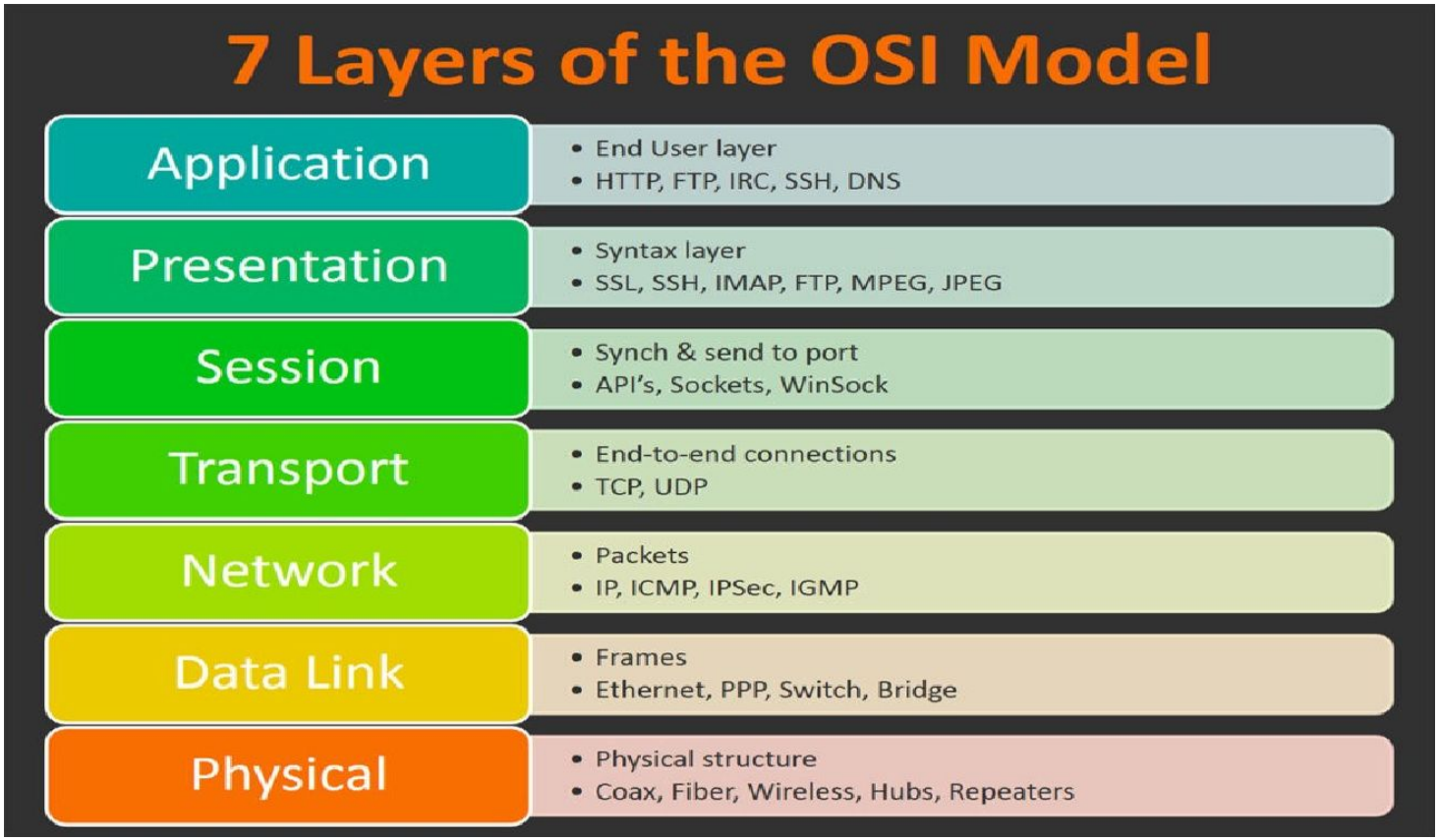
ESTD:1995

The OSI Reference Model



THE OSI MODEL

ESTD:1995



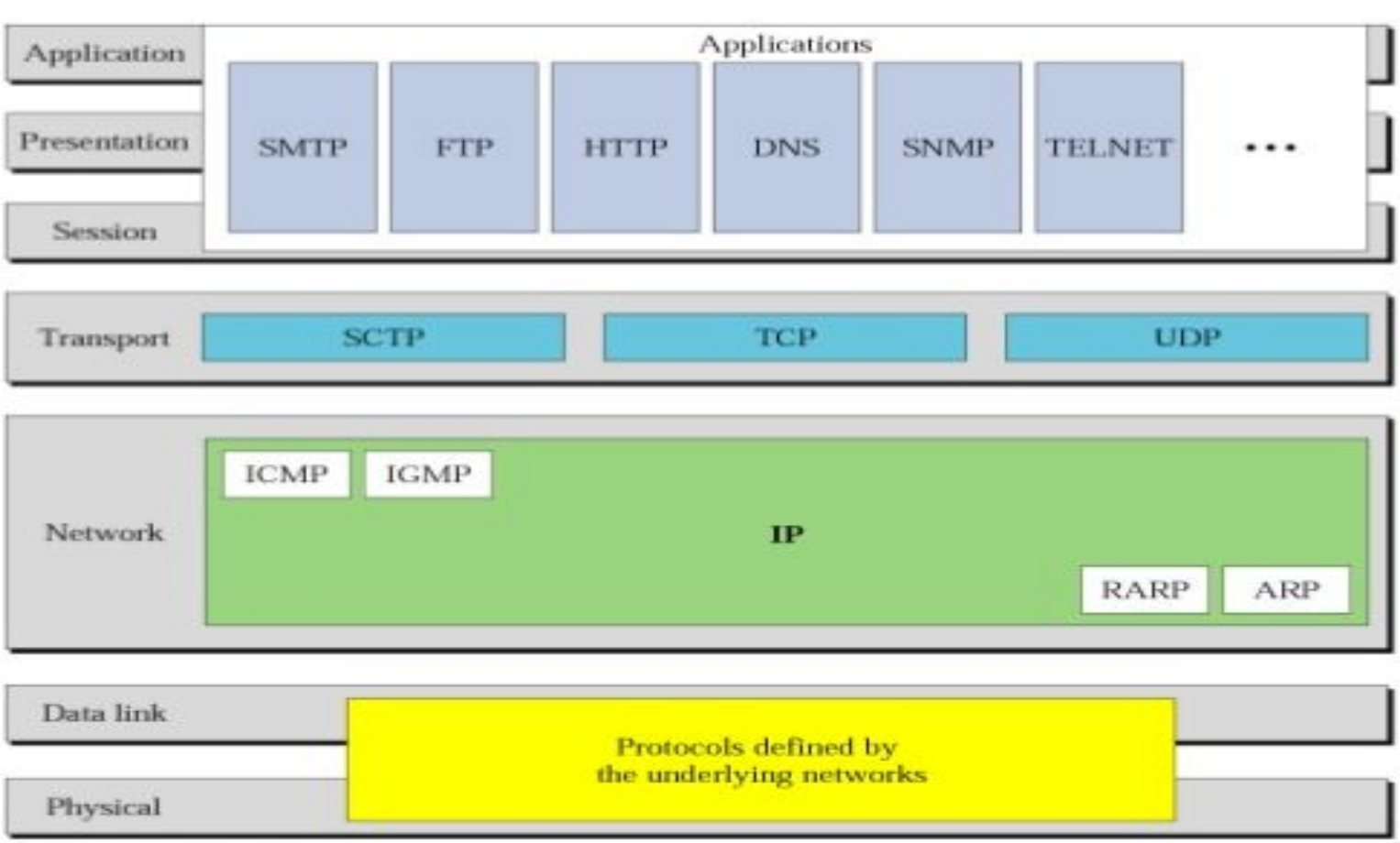
TCP/IP PROTOCOL SUITE

ESTD:1995

- The TCP/IP protocol suite was developed prior to the OSI model. Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application. However, when TCP/IP is compared to OSI, the host-to-network layer is equivalent to the combination of the physical and data link layers.
- The internet layer is equivalent to the network layer, and the application layer is roughly doing the job of the session, presentation, and application layers with the transport layer in TCP/IP taking care of part of the duties of the session layer.
- TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.

TCP/IP PROTOCOL SUITE

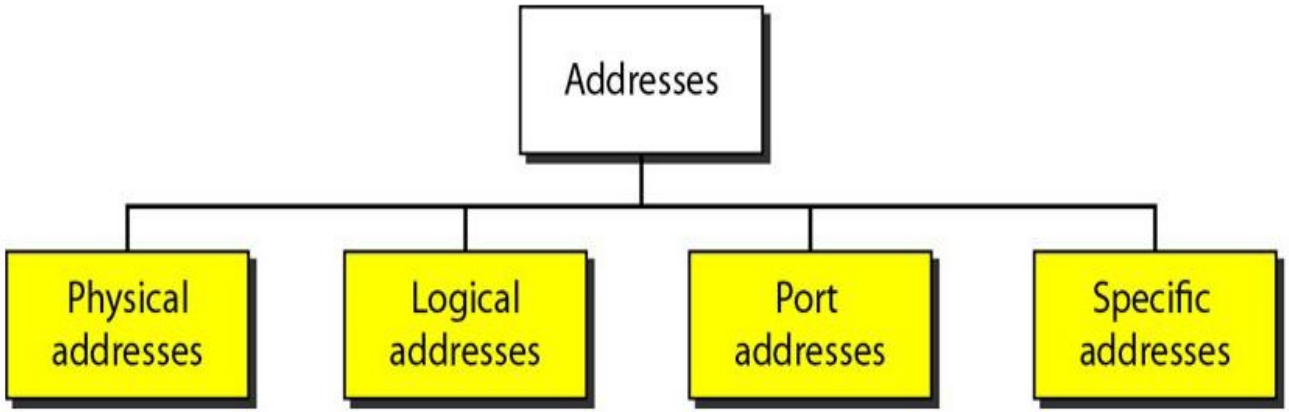
ESTD:1995



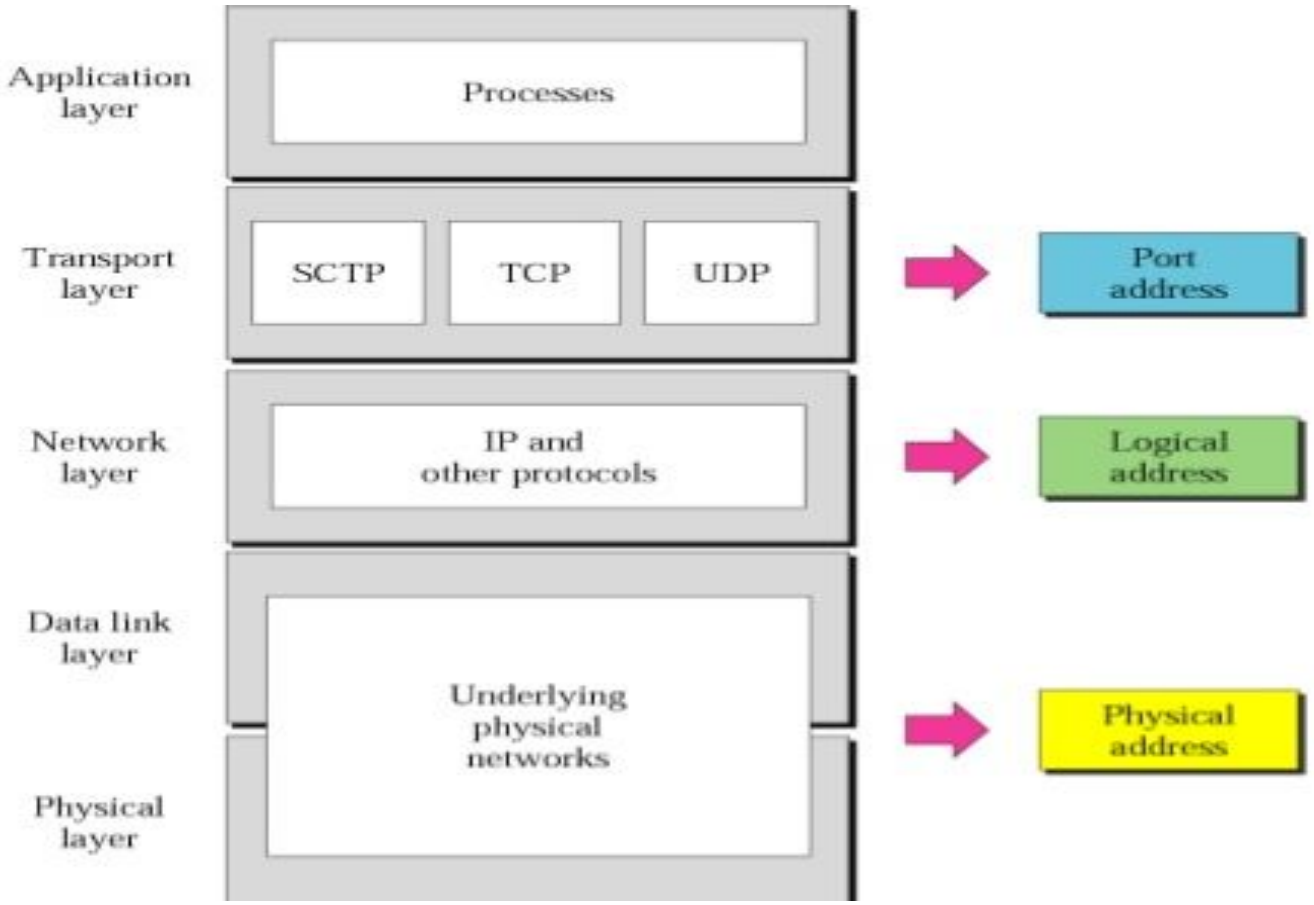
ADDRESSING

ESTD:1995

- Four levels of addresses are used in an internet employing the *TCP/IP* protocols: physical (link) addresses, logical (IP) addresses, port addresses, and specific addresses.



ADDRESSING


Application layer
Processes
Transport layer
SCTP
TCP
UDP
Port address
Network layer
IP and other protocols
Logical address
Data link layer
Underlying physical networks
Physical address
Physical layer

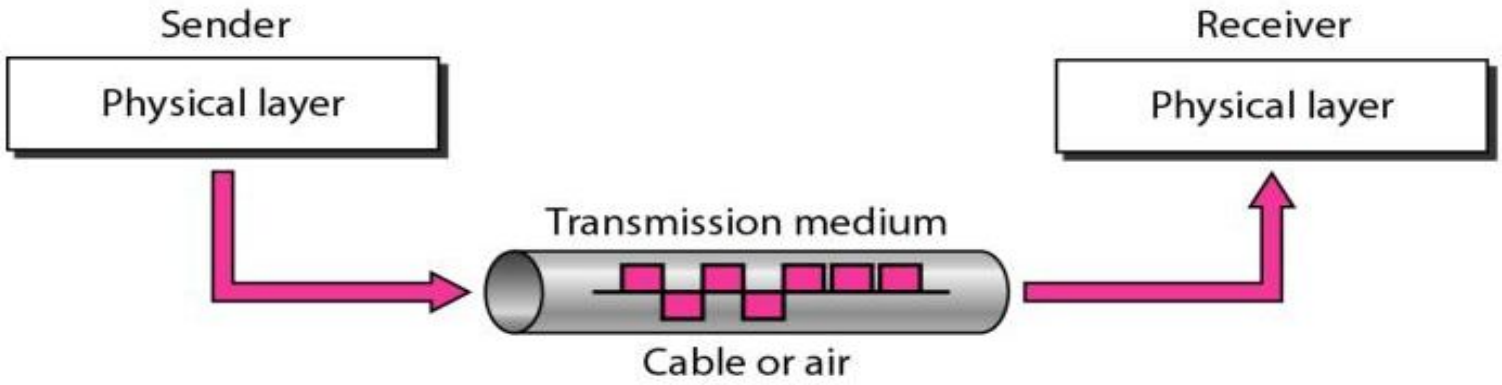
ADDRESSING

ESTD:1995

- **Physical Addresses**
  - The physical address known as the link address, is the address of a node as defined by its LAN or WAN. It is included in the frame used by the data link layer. It is the lowest-level address.
- **Logical Addresses**
  - Logical addresses are necessary for universal communications that are independent of underlying physical networks. Physical addresses are not adequate in an internetwork environment
- **Port Addresses**
  - Today, computers are devices that can run multiple processes at the same time. The end objective of Internet communication is a process communicating with another process.
- **Specific Addresses**
  - Some applications have user-friendly addresses that are designed for that specific address. Examples include the e-mail address (for example, munirajunaidu_v@vnrvjiet.in) and the Universal Resource Locator (URL) (for example, www.vnrvjiet.ac.in.).

Transmission Media

- A transmission **medium** carries information from a source to a destination. *For example, the transmission medium for two people having a dinner conversation is the air*.
- The transmission medium is usually free space, metallic cable, or fiber-optic cable. The information is usually a signal that is the result of a conversion of data from another form



Transmission Media

ESTD:1995

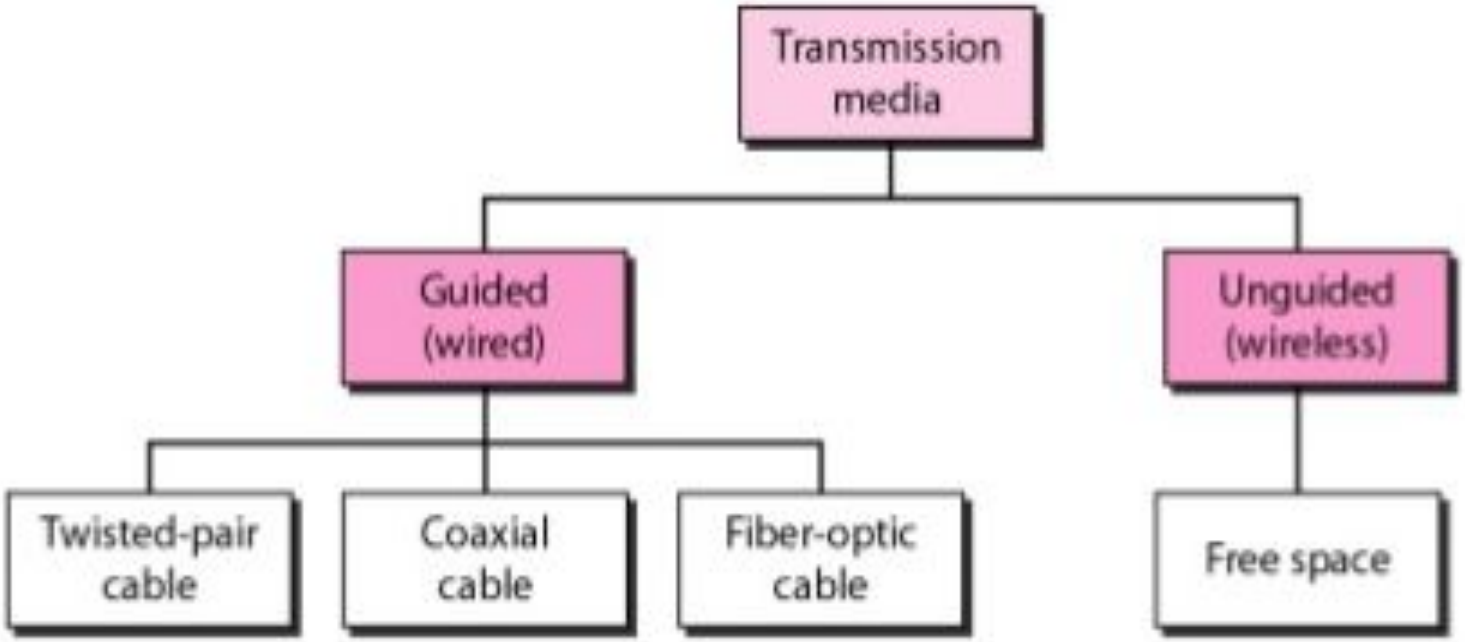
- The use of long-distance communication using electric signals started with the invention of the telegraph by Morse in the 19th century. Communication by telegraph was slow and dependent on a metallic medium.
- Wireless communication started in 1895 when Hertz was able to send high frequency signals. Later, Marconi devised a method to send telegraph-type messages over the Atlantic Ocean. Better metallic media have been invented (twisted pair and coaxial cables, for example).
- The use of optical fibers has increased the data rate incredibly. Free space (air, vacuum, and water) is used more efficiently due to the technologies (such as modulation and multiplexing) .
- Computers and other telecommunication devices use signals to represent data. These signals are transmitted from one device to another in the form of electromagnetic energy, which is propagated through transmission media.

Transmission Media

ESTD:1995

- **Some factors need to be considered for designing the transmission media:**
  - **Bandwidth:** The greater the bandwidth of a medium, the higher the data transmission rate of a signal.
  - **Transmission impairment:** When the received signal is not identical to the transmitted one due to the transmission impairment, the quality of the signals will get destroyed due to transmission impairment.
  - **Interference:** The process of disrupting a signal when it travels over a communication medium on the addition of some unwanted signal.
- **Causes of Transmission Impairment:**
  - **Attenuation:** Attenuation means the loss of energy, i.e., the strength of the signal decreases with increasing the distance which causes the loss of energy.
  - **Distortion:** Distortion occurs when there is a change in the shape of the signal. Each frequency component has its own propagation speed, so they reach at a different time which leads to the delay distortion.
  - **Noise:** When data is travelled over a transmission medium, some unwanted signal is added to it which creates the noise.

Classes of Transmission Media



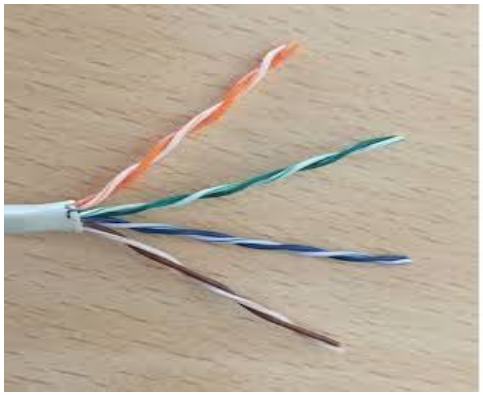
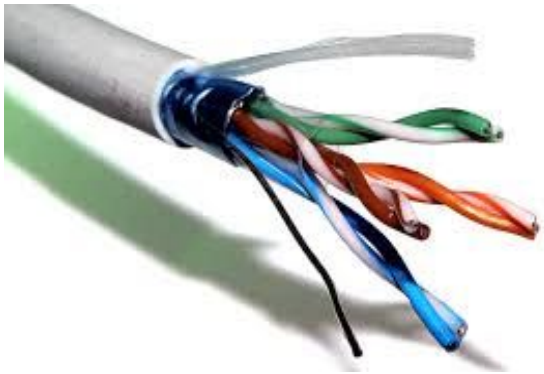
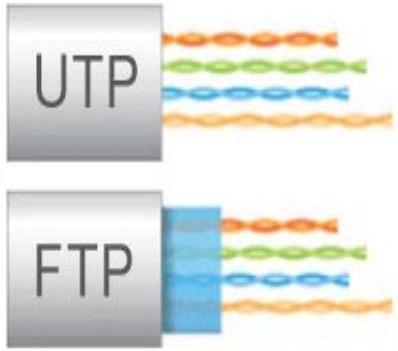
Classes of Transmission Media

- Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.
- A signal traveling along any of these media is directed and contained by the physical limits of the medium.
- Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current.
- Optical fiber is a cable that accepts and transports signals in the form of light.

Twisted Pair Cable

ESTD:1995

- Twisted pair cables were mainly invented for voice transmissions.
- Twisted pair is a widely used medium in networking because it's lighter, cheaper, more flexible, easy to install, and provides greater speeds than coaxial cables.
- Two types of twisted pair cables: the unshielded twisted pair (UTP) and the shielded twisted pair (STP).

		
Unshielded Twisted Pair Cable	**Shielded Twisted Pair Cable**	**UTP and STP Difference**

Twisted Pair Cable

ESTD:1995

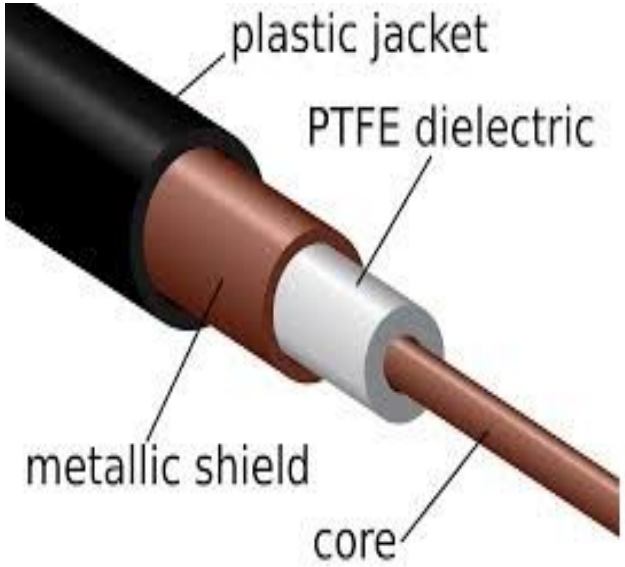
- An unshielded twisted pair is widely used in telecommunication. Following are the categories of the unshielded twisted pair cable:
  - **Category 1:** Category 1 is used for telephone lines that have low-speed data.
  - **Category 2:** It can support up to 4Mbps.
  - **Category 3:** It can support up to 16Mbps.
  - **Category 4:** It can support up to 20Mbps. Therefore, it can be used for long-distance communication.
  - **Category 5:** It can support up to 200Mbps.

Coaxial Cables

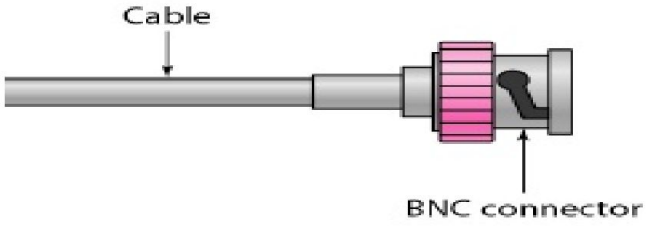
ESTD:1995

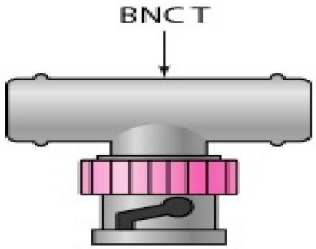
- The coaxial cables have a central copper conductor, surrounded by an insulating layer, a conducting shield, and the outermost plastic sheath.
- There are three insulation layers for the inner copper cable. There are two basic modes of data transmission in coaxial cables: baseband mode that has dedicated bandwidth, and broadband mode that has distributed cable bandwidth.
- Cable TV and analog televisions mainly use coaxial cables. Coaxial cables have better resistance to cross talk than twisted pair cables. The coaxial cables are used for long distance communication

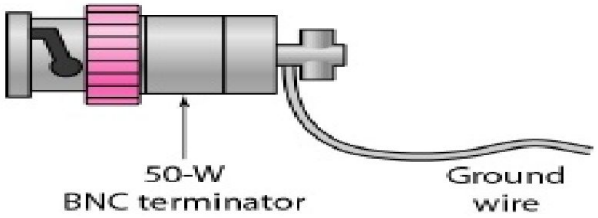
Coaxial Cables



1. The BNC connector is used to connect the end of the cable to a device, such as a TV set.
2. The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device.
3. The BNC terminator is used at the end of the cable to prevent the reflection of the signal.







Coaxial Cables



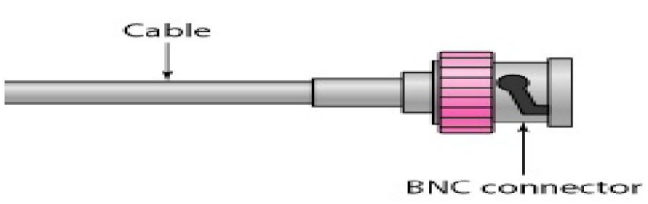
Bus Topology

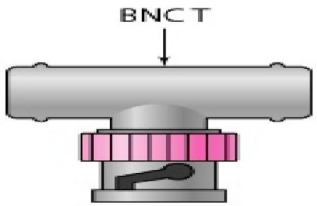


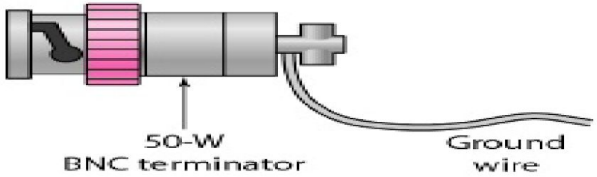
Terminator



T Connecter



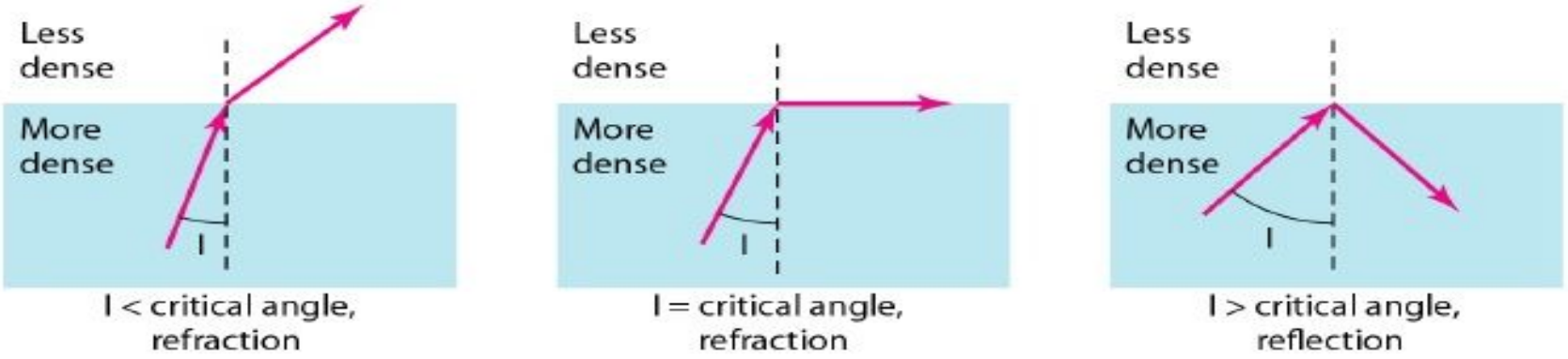




Fiber-Optic Cable

ESTD:1995

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.



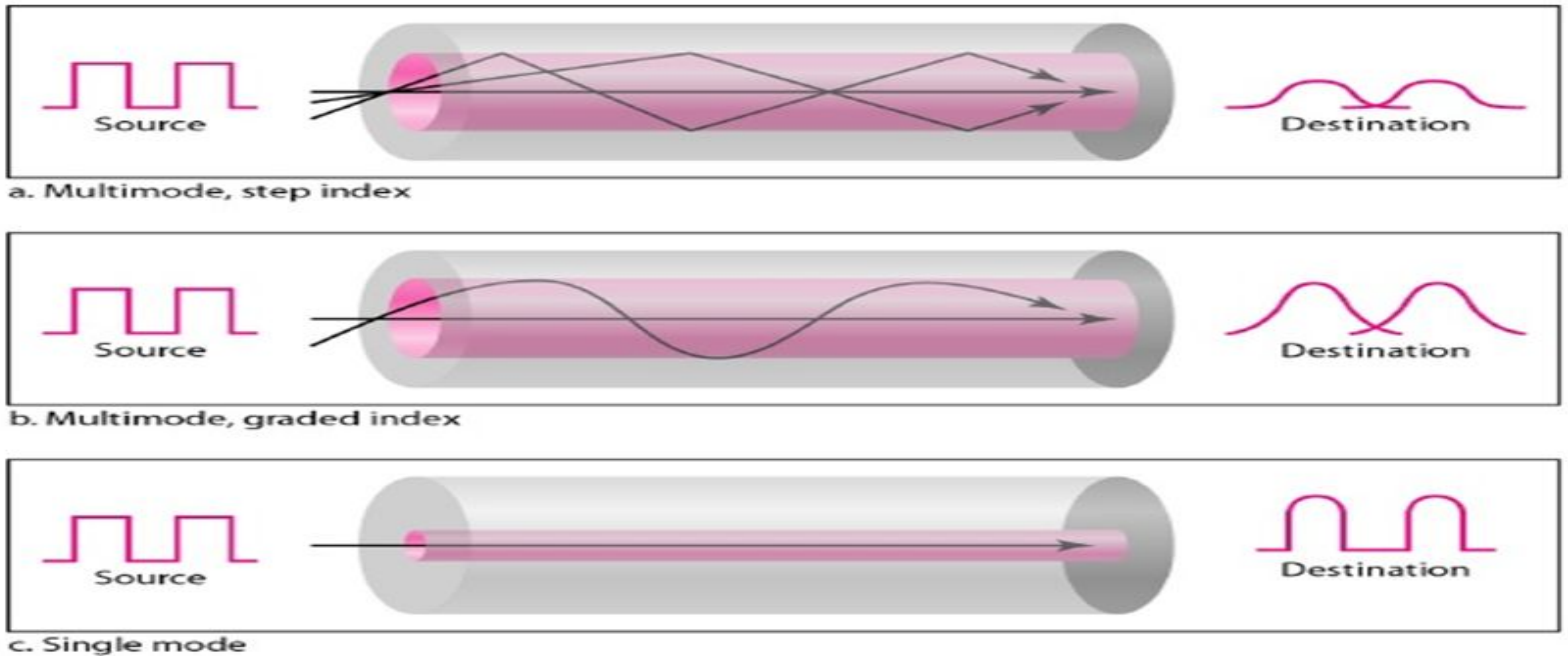
Propagation Modes

Current technology supports two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics. Multimode can be implemented in two forms: step-index or graded-index .

Fiber-Optic Cable

ESTD:1995

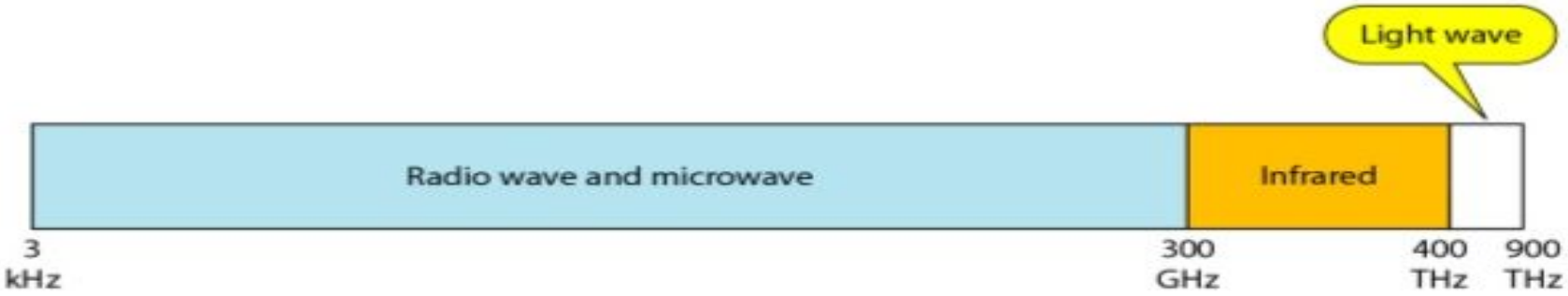
- **Multimode:** Here multiple beams from a light source move through the core in different paths. The beams that move within the cable depends on the structure of the core.
  - **multimode step-index fiber**
  - **multimode graded-index fiber**



UNGUIDED MEDIA: WIRELESS

ESTD:1995

- Unguided media transport electromagnetic waves without using a physical conductor often referred to as wireless communication. Signals are broadcasted through free space and thus are available to anyone who has a device capable of receiving them.
- The below figure shows the part of the electromagnetic spectrum, ranging from 3 kHz to 900 THz, used for wireless communication.

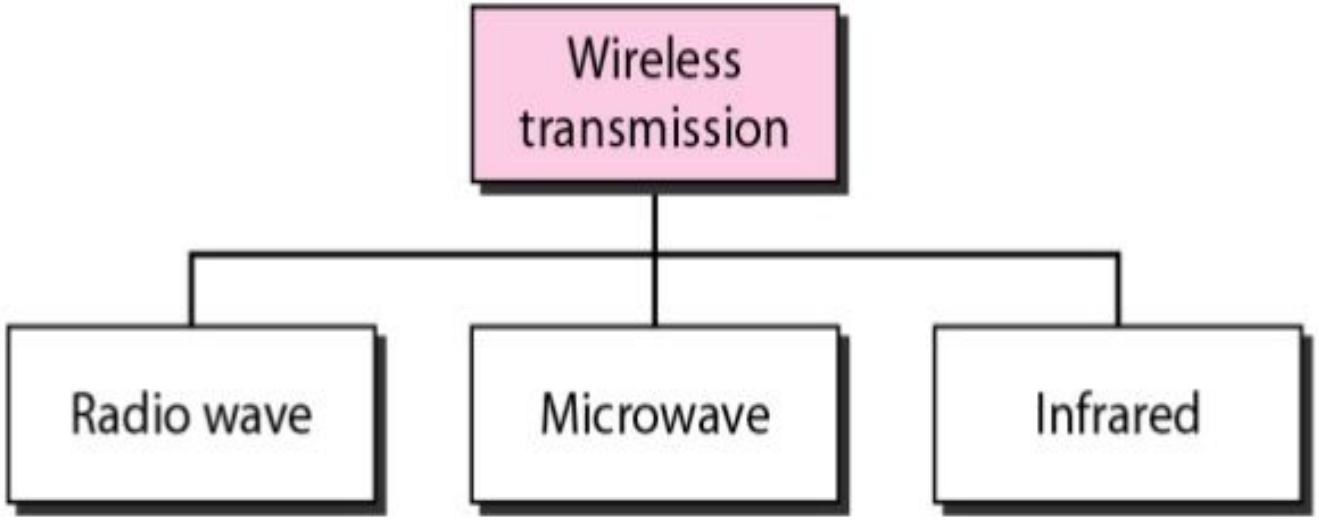


UNGUIDED MEDIA: WIRELESS

ESTD:1995

- Unguided signals can travel from the source to destination in several ways: ground propagation, sky propagation, and line-of-sight propagation.
  - **ground propagation**, radio waves travel through the lowest portion of the atmosphere and these low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet. Distance depends on the amount of power in the signal: The greater the power, the greater the distance.
  - **Sky propagation**, higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth. This type of transmission allows for greater distances with lower output power.
  - **line-or-sight propagation**, very high-frequency signals are transmitted in straight lines directly from antenna to antenna. Antennas must be directional, facing each other.

UNGUIDED MEDIA: WIRELESS



UNGUIDED MEDIA: WIRELESS

ESTD:1995

- **Radio Waves**
  - Electromagnetic waves called radio waves ranging in frequencies between **3 kHz and 1 GHz are normally called radio waves**; waves ranging in frequencies **between 1GHz and 300 GHz are called microwaves.**
  - ***Applications***
    - The omnidirectional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers. AM and FM radio, television etc.,.
- **Microwaves**
  - Microwaves are unidirectional. When an antenna transmits microwave waves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.

UNGUIDED MEDIA: WIRELESS

ESTD:1995

- **Microwave propagation is line-of-sight**. Antennas need to be in direct sight of each other, towers that are far apart need to be very tall. Repeaters are often needed for long distance communication.
  - Very high-frequency microwaves cannot penetrate walls which can be a disadvantage if receivers are inside buildings.
  - The microwave band is relatively wide, almost 299 GHz. Therefore wider sub-bands can be assigned, and a high data rate is possible
  - Use of certain portions of the band requires permission from authorities.
- Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used : the parabolic dish and the horn.
  - **Applications**
    - *V*ery useful when unicast (one-to-one) communication is needed between the sender and the receiver. They are used in cellular phones.

UNGUIDED MEDIA: WIRELESS

ESTD:1995

Infrared

- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication.
- Infrared waves, having high frequencies, cannot penetrate walls. Prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room.
- When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. However, this same characteristic makes infrared signals useless for long-range communication.

Multiplexing

ESTD:1995

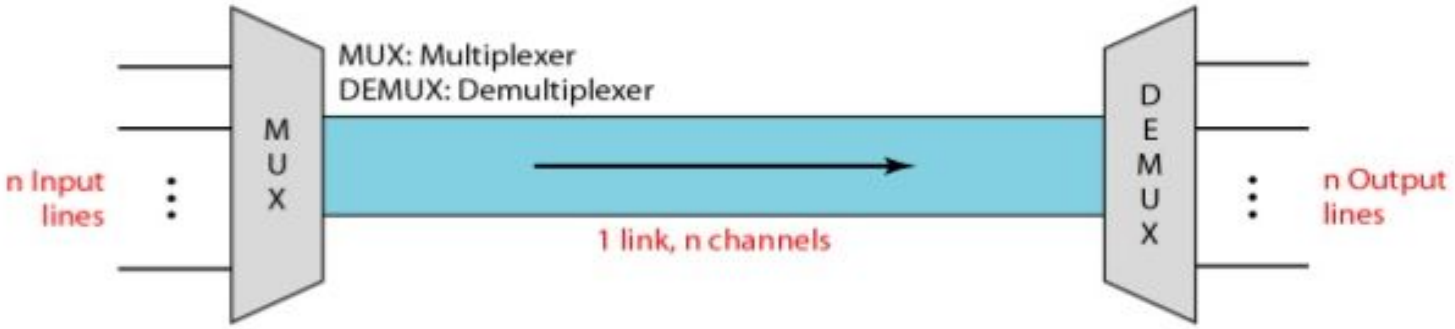
- Links with limited bandwidths.
- The effective utilization of these bandwidths is one of the main challenges of electronic communications.
- There is a need to combine several low-bandwidth channels to make use of one channel with a larger bandwidth.
- There is a need to expand the bandwidth of a channel to achieve goals such as privacy and anti-jamming.
- There are two broad categories of bandwidth utilization: multiplexing and spreading.
- In multiplexing, the goal is efficiency.
- In spreading, the goals are privacy and anti-jamming, the bandwidth of the channel is expanded to insert redundancy, which is necessary to achieve these goals

Multiplexing

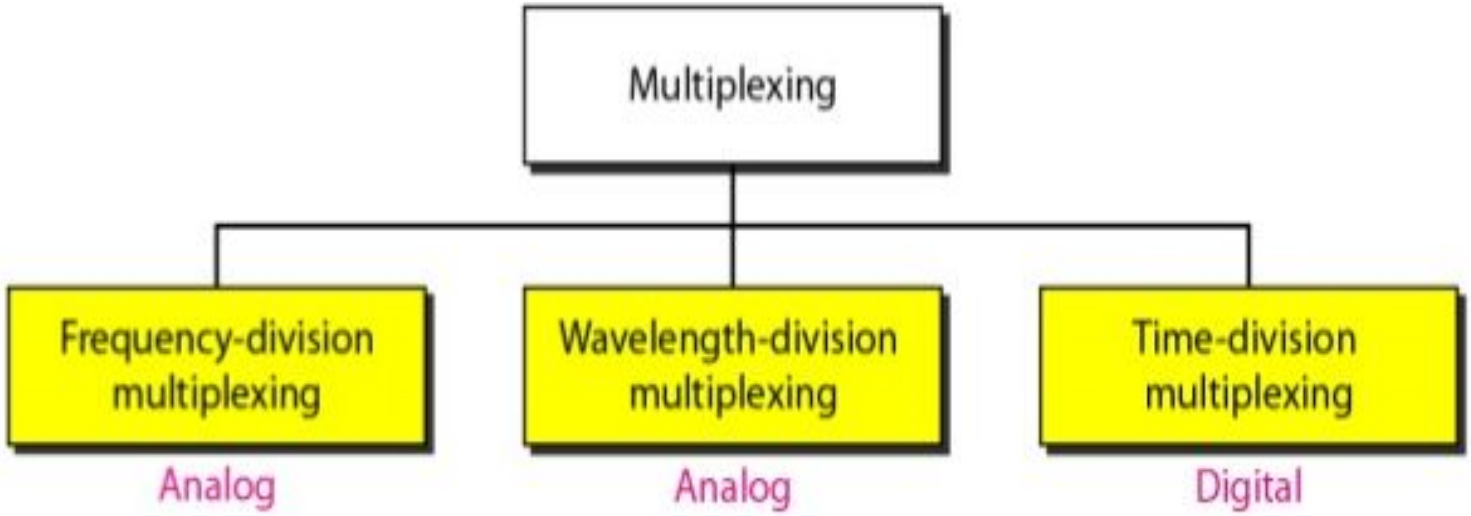
In a multiplexed system, *n* lines share the bandwidth of one link. Figure 6.1 shows the basic format of a multiplexed system.

The lines on the left direct their transmission streams to a multiplexer (MUX), which combines them into a single stream (many-to-one).

At the receiving end, that stream is fed into a de-multiplexer (DEMUX), which separates the stream back into its component transmissions (one-to-many) and directs them to their corresponding lines.



Categories of multiplexing

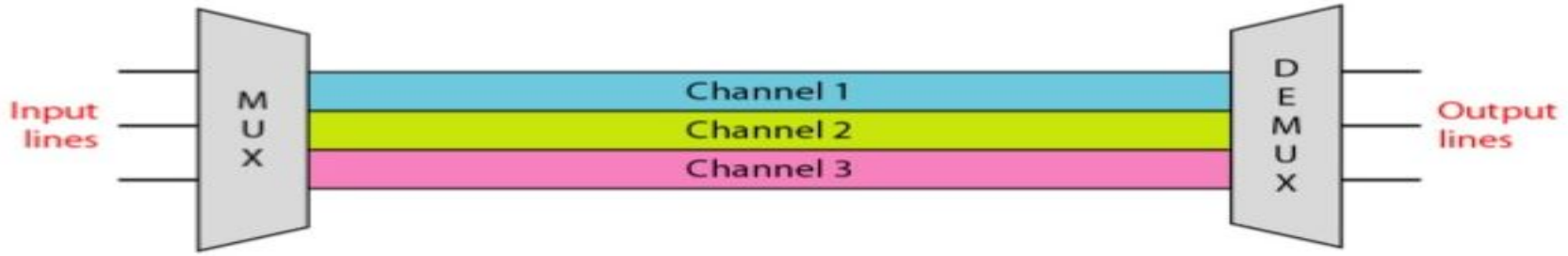


Frequency-Division Multiplexing

ESTD:1995

- Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.
- In FOM, signals generated by each sending device modulate different carrier frequencies. These modulated signals are then combined into a single composite signal that can be transported by the link.
- Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal.
- These bandwidth ranges are the channels through which the various signals travel. Channels can be separated by strips of unused bandwidth-guard bands-to prevent signals from overlapping.
- In addition, carrier frequencies must not interfere with the original data frequencies.

Frequency-Division Multiplexing



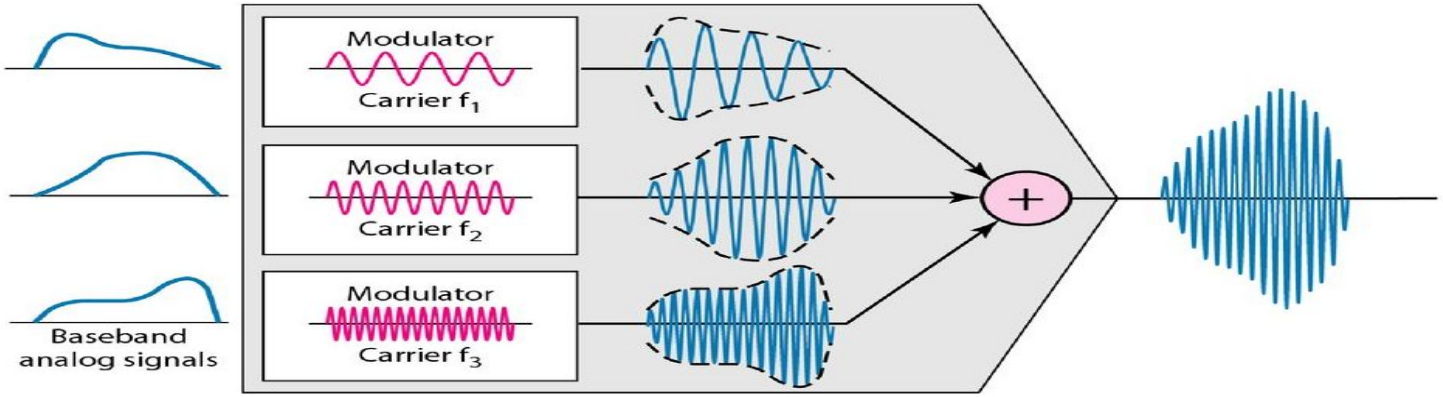
- FDM is considered to be an analog multiplexing technique; however, this does not mean that FDM cannot be used to combine sources sending digital signals.
- A digital signal can be converted to an analog signal before FDM is used to multiplex them

Frequency-Division Multiplexing

ESTD:1995

Multiplexing Process

- Each source generates a signal of a similar frequency range. Inside the multiplexer, these similar signals modulates different carrier frequencies *(f1,f2,f3).*
- The resulting modulated signals are then combined into a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.

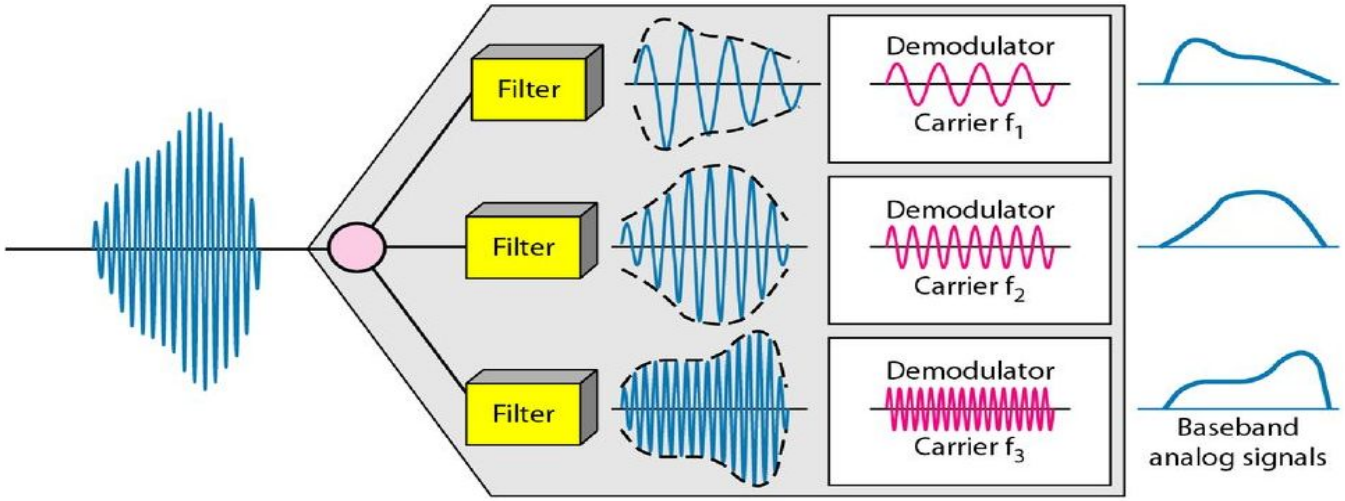


Frequency-Division Multiplexing

ESTD:1995

Demultiplexing Process

- The de-multiplexer uses a series of filters to decompose the multiplexed signal into its constituent component signals.
- The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the output lines.
- The below figure shows the conceptual illustration of de-multiplexing process.

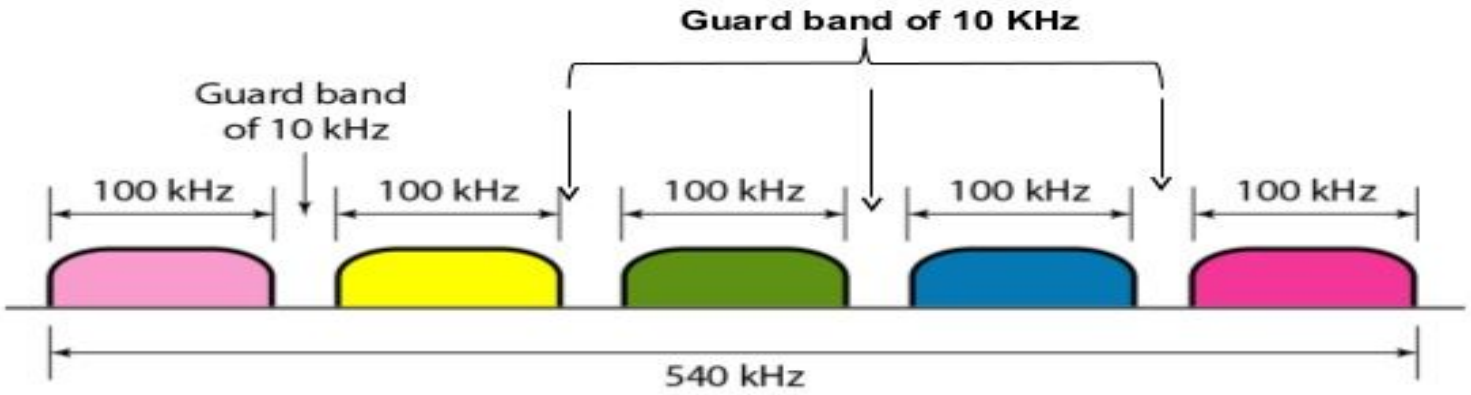


Exercise problems on FDM

ESTD:1995

Example : Five channels, each with a 10 a-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10kHz between the channels to prevent interference?

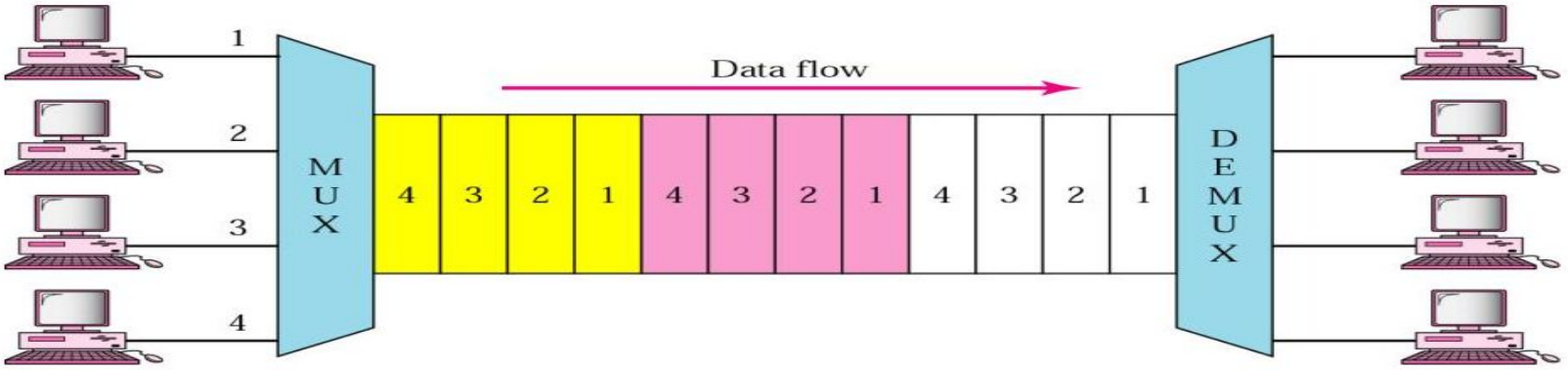
Solution: For five channels, we need at least four guard bands. This means that the required bandwidth is at least 5 x 100 + 4 x 10 =540 kHz, as shown in the below figure



Synchronous Time-Division Multiplexing

ESTD:1995

- Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a line. Instead of sharing a portion of the bandwidth as in FDM, time is shared.
- Each connection occupies a portion of time in the link. The below figure gives a conceptual view of TDM. Note that the same link is used as in FDM; here, however, the link is shown sectioned by time rather than by frequency.
- In the figure, portions of signals 1,2,3, and 4 occupy the link sequentially.

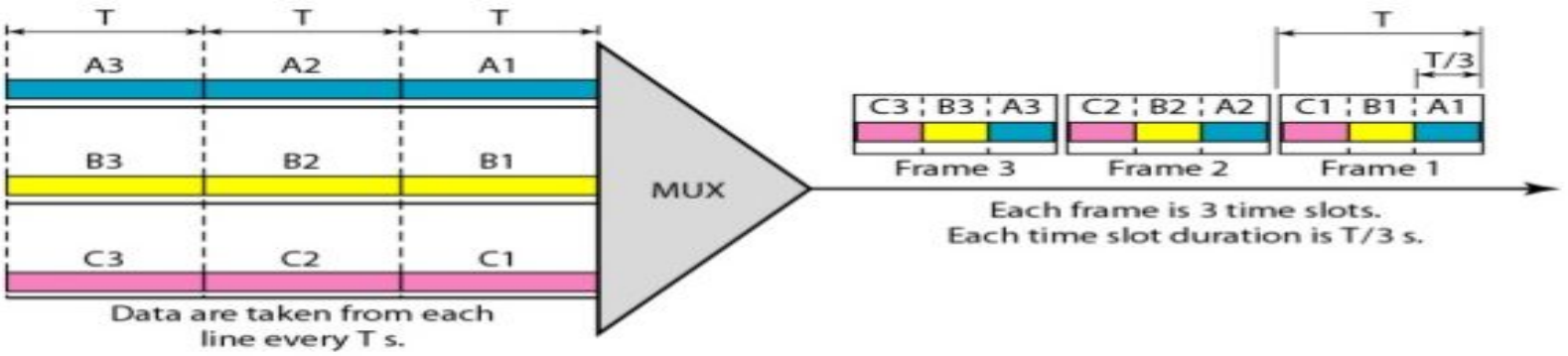


Synchronous Time-Division Multiplexing

ESTD:1995

Time Slots and Frames

- In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot.
- A unit can be 1 bit, one character, or one block of data.
- Each input unit becomes one output unit and occupies one output time slot. However, the duration of an output time slot is *n* times shorter than the duration of an input time slot. The below figure shows an example of synchronous TDM where *n* is 3.



Synchronous Time-Division Multiplexing

ESTD:1995

Example 6.5

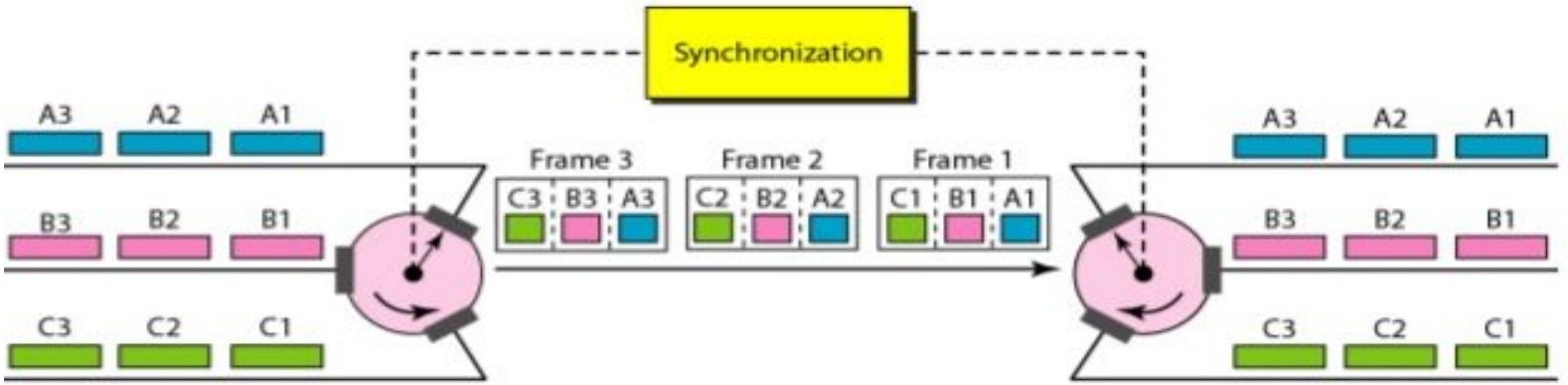
If the data rate for each input connection is 3 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

Solution

- The data rate of each input connection is 1 kbps. This means that the bit duration is 111000 or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).
- The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is 1/3 ms.
- Each frame carries three output time slots. So the duration of a frame is 3 x 113 ms, or 1ms.The duration of a frame is the same as the duration of an input unit.

TDM: Interleaving

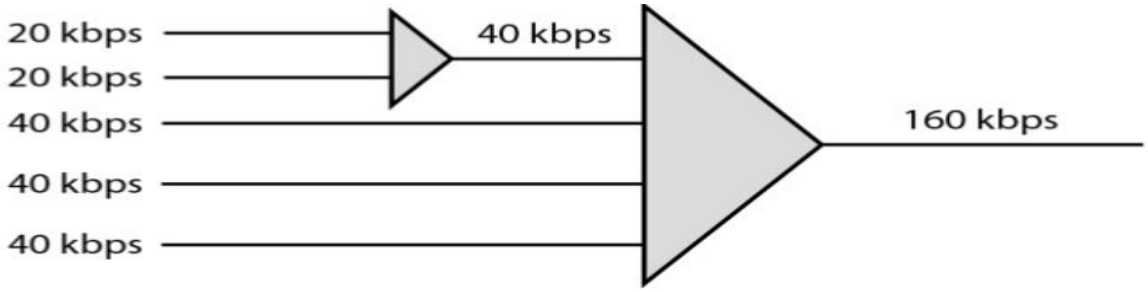
- TDM can be visualized as two fast-rotating switches, one on the multiplexing side and the other on the demultiplexing side. The switches are synchronized and rotate at the same speed, but in opposite directions.
- On the multiplexing side, as the switch opens in front of a connection, that connection has the opportunity to send a unit onto the path. This process is called **interleaving.** On the demultiplexing side, as the switch opens in front of a connection, that connection has the opportunity to receive a unit from the path.



TDM

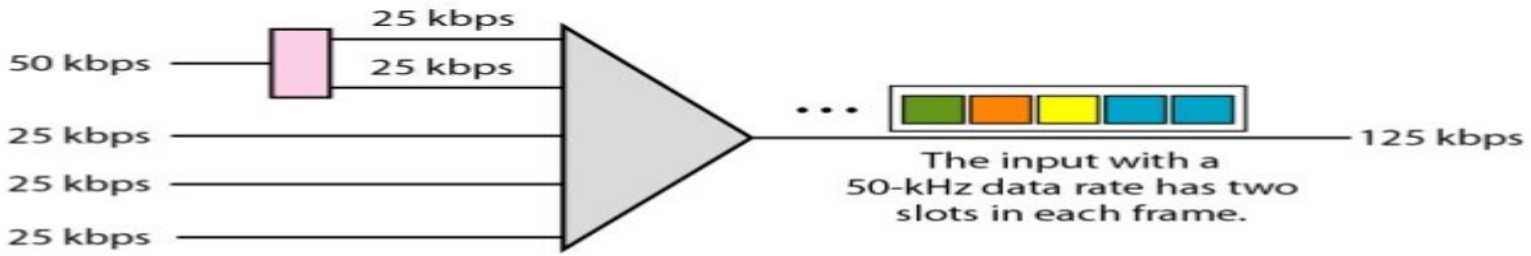
ESTD:1995

- One problem with TDM is how to handle a disparity in the input data rates. If data rates are not the same, three strategies, or a combination of them, can be used. The three strategies are called as **multilevel multiplexing, multiple-slot allocation,** and **pulse stuffing.**
  - **Multilevel Multiplexing** Multilevel multiplexing is a technique used when the data rate of an input line is a multiple of others.

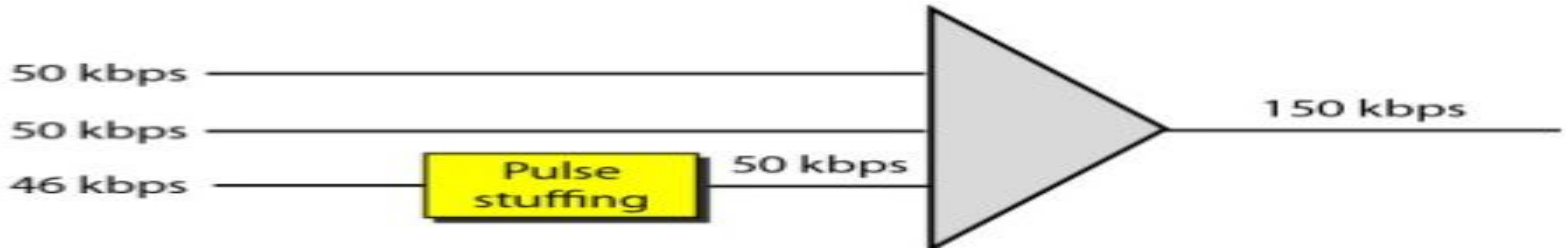


Multiple-Slot Allocation: Sometimes it is more efficient to allot more than one slot in a frame to a single input line. For example, we might have an input line that has a data rate that is a multiple of another input.

TDM



Pulse Stuffing Sometimes the bit rates of sources are not multiple integers of each other. Therefore, neither of the above two techniques can be applied. One solution is to make the highest input data rate the dominant data rate and then add dummy bits to the input lines with lower rates. This will increase their rates. This technique is called pulse stuffing, bit padding, or bit stuffing.

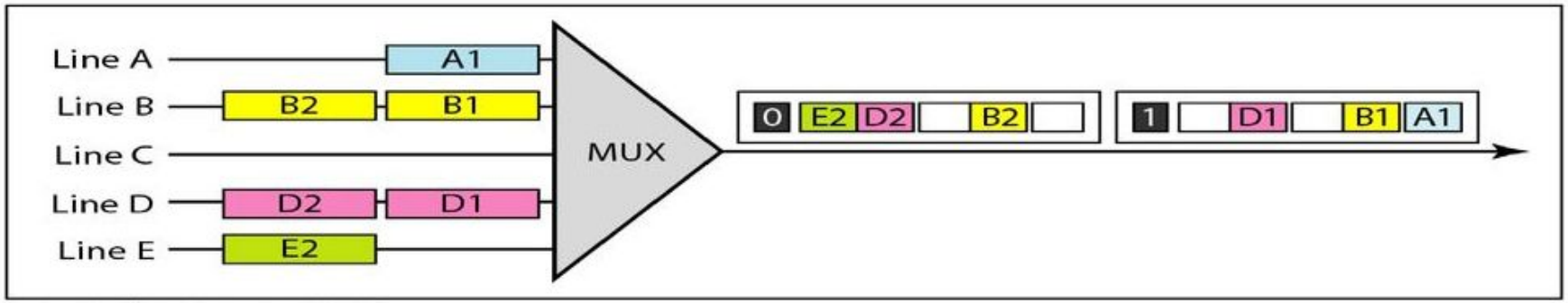


Statistical Time-Division Multiplexing

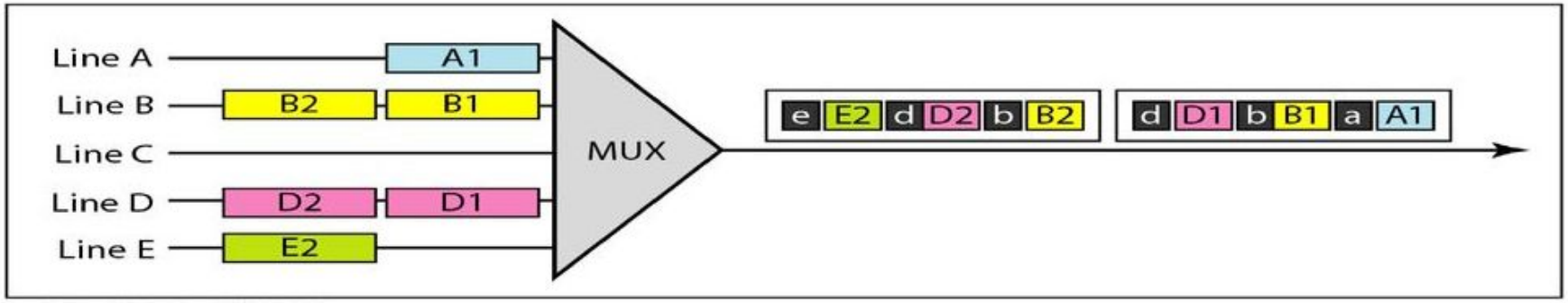
ESTD:1995

- In synchronous TDM, each input has a reserved slot in the output frame. This can be inefficient if some input lines have no data to send.
- In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency.
- Only when an input line has a slot's worth of data to send is it given a slot in the output frame.
- In statistical multiplexing, the number of slots in each frame is less than the number of input lines.
- The multiplexer checks each input line in round-robin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line.

Statistical Time-Division Multiplexing



a. Synchronous TDM



b. Statistical TDM

Statistical Time-Division Multiplexing

ESTD:1995

- In statistical TDM, a slot needs to carry data as well as the address of the destination.
- In synchronous TDM, there is no need for addressing, synchronization and preassigned relationships between the inputs and outputs serve as an address.
- In statistical multiplexing, there is no fixed relationship between the inputs and outputs because there are no preassigned or reserved slots.
- There is a need to include the address of the receiver inside each slot to show where it is to be delivered.
- ***Slot Size***
  - Since a slot carries both data and an address in statistical TDM, the ratio of the data size to address size must be reasonable to make transmission efficient. For example, it would be inefficient to send 1 bit per slot as data when the address is 3 bits.(an overhead of 300 percent)

Statistical Time-Division Multiplexing

ESTD:1995

- In statistical TDM, a slot needs to carry data as well as the address of the destination.
- In synchronous TDM, there is no need for addressing, synchronization and preassigned relationships between the inputs and outputs serve as an address.
- In statistical multiplexing, there is no fixed relationship between the inputs and outputs because there are no preassigned or reserved slots.
- There is a need to include the address of the receiver inside each slot to show where it is to be delivered.
- ***Slot Size***
  - Since a slot carries both data and an address in statistical TDM, the ratio of the data size to address size must be reasonable to make transmission efficient. For example, it would be inefficient to send 1 bit per slot as data when the address is 3 bits.(an overhead of 300 percent)

TDM

ESTD:1995

- **No Synchronization Bit**
  - There is another difference between synchronous and statistical TDM, but this time it is at the frame level. The frames in statistical TDM need not be synchronized, so we do not need synchronization bits.
- **Bandwidth**
  - In statistical TDM, the capacity of the link is normally less than the sum of the capacities of each channel. The designers of statistical TDM define the capacity of the link based on the statistics of the load for each channel. If on average only *x* percent of the input slots are filled, the capacity of the link reflects this. Of course, during peak times, some slots need to wait.

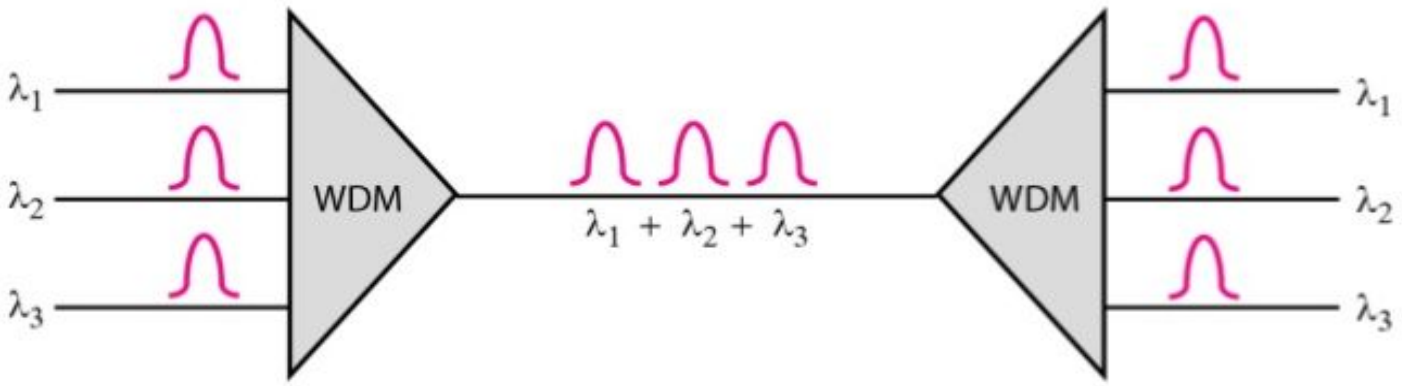
WDM

ESTD:1995

- Wavelength-division multiplexing (WDM) is designed to use the high-data-rate capability of fiber-optic cable.
- The optical fiber data rate is higher than the data rate of metallic transmission cable. Using a fiber-optic cable for one single line wastes the available bandwidth. Multiplexing allows us to combine several lines into one.
- WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels.
- combining different signals of different wavlengths.

WDM

- The below figure gives a conceptual view of a WDM multiplexer and demultiplexer. Very narrow bands of light from different sources are combined to make a wider band of light. At the receiver, the signals are separated by the demultiplexer.



WDM

- The basic idea of WDM technology is to combine multiple light sources into one single light at the multiplexer and do the reverse at the demultiplexer. The combining and splitting of light sources are easily handled by a prism.
- One application of WDM is the SONET network in which multiple optical fiber lines are multiplexed and demultiplexed.
- A new method, called dense WDM (DWDM), can multiplex a very large number of channels by spacing channels very close to one another. It achieves even greater efficiency

SPREAD SPECTRUM

- Multiplexing combines signals from several sources to achieve bandwidth efficiency: the available bandwidth of a link is divided between the sources. In spread spectrum, the signals are combined from different sources to fit into a larger bandwidth.
- Spread spectrum is designed to be used in wireless applications (LANs and WANs). In these types of applications, there are some concerns that outweigh bandwidth efficiency. In wireless applications, all stations use air (or a vacuum) as the medium for communication. Stations must be able to share this medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder (in military operations, for example).

SPREAD SPECTRUM

ESTD:1995

- To achieve these goals, spread spectrum techniques add redundancy; they spread the original spectrum needed for each station. If the required bandwidth for each station is *B,* spread spectrum expands it to B_{SS} such that $B_{SS} \gg B$. The expanded bandwidth allows the source to wrap its message in a protective envelope for a more secure transmission.

Spread spectrum achieves its goals through two principles:

1. The bandwidth allocated to each station needs to be, by far, larger than what is needed which allows redundancy.
2. The expanding of the original bandwidth *B* to the bandwidth B_{SS} must be done by a process that is independent of the original signal. In other words, the spreading process occurs after the signal is created by the source.

SPREAD SPECTRUM

ESTD:1995

- There are two techniques to spread the bandwidth: frequency hopping spread spectrum (FHSS) and direct sequence spread spectrum (DSSS).

Frequency Hopped Spread Spectrum(FHSS)

- In frequency hopping technique, where the users are made to change the frequencies of usage, from one to another in a specified time interval, hence called as **frequency hopping**.
- For example, a frequency was allotted to sender 1 for a particular period of time. Now, after a while, sender 1 hops to the other frequency and sender 2 uses the first frequency, which was previously used by sender 1. This is called as **frequency reuse**.
- The frequencies of the data are hopped from one to another in order to provide a secure transmission. The amount of time spent on each frequency hop is called as **Dwell time**.

SPREAD SPECTRUM

ESTD:1995

- Whenever a user wants to send data using this DSSS technique, each and every bit of the user data is multiplied by a secret code, called as **chipping code**.
- This chipping code is the spreading code which is multiplied with the original message and transmitted.
- The receiver uses the same code to retrieve the original message.

Advantages of Spread Spectrum

- Crosstalk elimination ,Better output with data integrity ,Reduced effect of multipath fading
- Better security ,Reduction in noise, Co-existence with other systems, Longer operative distances
- Hard to detect, Not easy to demodulate/decode, Difficult to jam the signals, Although spread spectrum techniques were originally designed for military uses, they are now being used widely for commercial purpose.

SPREAD SPECTRUM

ESTD:1995

FHSS	DSSS / CDMA
Multiple frequencies are used	Single frequency is used
Hard to find the user's frequency at any instant of time	User frequency, once allotted is always the same
Frequency reuse is allowed	Frequency reuse is not allowed
Sender need not wait	Sender has to wait if the spectrum is busy
Power strength of the signal is high	Power strength of the signal is low
Stronger and penetrates through the obstacles	It is weaker compared to FHSS

SPREAD SPECTRUM

ESTD:1995

It is never affected by interference	It can be affected by interference
It is cheaper	It is expensive
This is the commonly used technique	This technique is not frequently used

WIT and WIL Scenario

Practice Questions

ESTD:1995

1. What are the three criteria necessary for an effective and efficient network?(2M)
2. What are the three fundamental characteristics determine the effectiveness of the data communication system?(2M)
3. Explain the ISO-OSI model of computer network with a neat diagram.(16M)
4. Describe the goals of multiplexing?(8M)
5. Distinguish between synchronous and statistical TDM.(8M)
6. How do guided media differ from unguided media?
7. What is the purpose of cladding in an optical fiber?
8. How does sky propagation differ from line-of-sight propagation?
9. What is the difference between omnidirectional waves and unidirectional waves?

References

- http://highered.mheducation.com/sites/0072967757/student_view0/chapter1/multiple_choice_quiz.html
- http://citengg.blogspot.com/p/behrouz-forouzancomputer-networks4th.html
- https://study.com/academy/lesson/transmission-basics-in-networking-media-characteristics-connector-types.html
- https://www.geeksforgeeks.org/types-transmission-media/
- https://www.javatpoint.com/multiplexing-in-computer-network
- https://www.tutorialspoint.com/data_communication_computer_network/physical_layer_multiplexing.htm
- https://www.computernetworkingnotes.com/networking-tutorials/multiplexing-and-demultiplexing-explained-with-types.html

ESTD:1995

ASSIGNMENT QUESTIONS

1.	Explain in detail about ISO OSI reference model	CO1
2.	Explain different layers and their functionalities in TCP/IP Model.	CO1
3.	a) Discuss in detail about the LAN and WAN. b)Differentiate between Wired LAN and Wireless LANs	CO1 CO1
4.	Elicit types of transmission media with their merits and demerits	CO1
5.	Explain Frequency division and Time division multiplexing in detail.	CO1

ESTD:1995

THANK YOU